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Review

Application of ultrasonic vibrations in welding and metal processing: A status review

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ABSTRACT

This review paper aims to introduce systematic in-depth analysis of diversified aspects of ultrasonic application in metal joining and processing including its limitations, future prospects and assessments. Allied welding and metal processing technologies employing ultrasonic vibrations either as the primary source to accomplish the intended operation or as an assistant source to improve the operation efficiency and product quality are classified and discussed. The detailed state-of the art, experimentation and progresses of the ultrasonic vibrations and its applications in the above areas are comprehensively examined, evaluated and presented for exhaustive understandings of its physical mechanism. The ultrasonic vibrations assisted processes claim several advantages such as improved mechanical properties, enlarged process window, higher heat generation, better material mixing and enhanced turbulence etc. over the conventional processes which lead to the remarkable and globalized applications of ultrasonic vibrations in the welding and its allied areas in the present century.

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1. Introduction

Modern era is on the brink of a serious energy crisis because on one hand the non-renewable energy sources are fast draining and on the other hand the renewable energy is still underdeveloped to meet the ever-growing energy demands. At this intricate stage, any action of saving energy is considered valuable from various aspects. Application of ultrasonic vibration has been proving to be indispensable in various diverse areas such as modern medicine and medical diagnostics, chemical and food processing, metal joining and processing etc., even by letting alone the direct service of ultrasound in highly sensitive sectors such as marine, safety and defence sectors. From a materials engineer's perspective, ultrasonic vibrations adds two main advantages to the material processing: lowers the energy consumption and the cost of the process. Ultrasonic application in metal working dates back to the early decades of 20th century. Different manufacturing processes such as machining, forming, joining, etc. are assisted with ultrasonic vibrations to improve the properties of the finished products. These ultrasonic vibrations themselves are the physical vibrations of the molecules in the medium in which the sound is traveling. The frequency range from ~20 kHz up to several gigahertz is usually termed as ultrasonic range. The effect of ultrasonic vibrations over the metals and alloys are mainly employed for manufacturing purposes. The outcome of ultrasonic integration in metal processing is sufficiently available in the literature [1–12]. Many researchers are exercising on inventing processes to facilitate acoustic softening during material processing. The effect of ultrasonic energy is similar to thermal softening; however, experimental results reveal that the ultrasonic energy required to produce an identical amount of softening is 10^7 times less than the required thermal energy [1,6]. Several researchers in their research work used ultrasonic vibrations for metal working and concluded that it is capable of reducing magnitude of force required to a large extent. Ultrasonic is the preferred method of metalworking in today's competitive world. Ultrasonic claims some advantages over others as follow:

(1) Ultrasonic process is clean, reliable, consistent and affordable.
(2) It needs no pre weld or post weld operations.
(3) It uses no consumables of any nature and is user and environment friendly, rapid and repeatable.
(4) Ultrasonic eliminates subjective elements in the welding process, ensuring consistent quality.
(5) The consumption of energy is perhaps the lowest by any other process [13].
(6) Set up is quick and easy. Its versatility enables change over from one set-up to another within a few minutes.
(7) Tooling has long life, and need little or no maintenance [13].

It is also termed as a green technology owing to the reason that it has an applicability for a wide spectrum of processes without any harm to the environment and human beings. This technology is primarily associated with the usage of high intensity elastic waves which were employed to cause changes in the treated media by the adequate exploitation of the nonlinear phenomena associated to the high amplitudes such as radiation pressure, wave distortion, streaming, and cavitation in liquids and dislocation in solids [14]. This leads to activation of different mechanisms such as agitation, diffusion, interface instabilities, friction reduction, localised heating, and mechanical rupture etc. [14]. As pointed out earlier, ultrasonic vibrations have been applied in various fields ranging from medical diagnostics to chemistry to materials joining and processing. In these different fields, the purpose of ultrasonic application is either primary or secondary. The primary purposes of ultrasonic application involve areas on medical diagnostics, food processing and ultrasonic welding etc. where ultrasonic vibrations play the main role in the process accomplishment. The secondary purposes involve application of ultrasonic vibrations to provide assistance to a desired process to improve the process effectiveness. For instance, ultrasonic application for both primary and secondary purposes has become widespread in various metal working processes. Few papers are available in literature that reviewed the ultrasonic application in machining of metals [15–20]. However, ultrasonic machining (USM) is only a narrow field of manufacturing processes while the ultrasonic vibrations have been widely employed in other areas of metal joining and processing such as in metal forming, fusion welding and solid state welding etc. Therefore, there is an urgent need to compile and critically analyse the state-of-art of the concepts, methods of ultrasonic application, advantages and limitations in various metal joining and other manufacturing processes for quick reference. This paper is an attempt to provide a detailed review that comprehensively covers all the neglected areas of ultrasonic application in the welding and its allied domains.

2. Instrumentation of ultrasonic system

A typical ultrasonic system is a sequential assembly of four major components: generator, transducer, booster, and horn. When integrated to each other, generates and subsequently transfers ultrasonic vibrations to a desired location of metal during a manufacturing process. Conversely, these components are integrated in a predesignated manner to achieve a desired ultrasonic power transfer efficiency. Efficient transfer of ultrasonic power is extremely vital and depends upon the elaborated and in depth design of the ultrasonic components. The design and selection of component materials depend upon the intended area of ultrasonic application. Detailed knowledge of each component is vital for robust designs and desired results.

2.1. Ultrasonic generator

Ultrasonic generator is a solid electronic box which is fed in with 220 V and 50 Hz from the mains as inputs and transforms the input into 800–1000 V and 20 kHz electrical supply. The anatomy of the generator is shown in Fig. 1. The ultrasonic generator majorly constitutes an electronic programmer, protection circuits and a microprocessor. If present, all the above generator parts are housed in a power supply cabinet. Power rating of the generator is based on the area of application and varies from plastic welding to metal welding. Two types of ultrasonic generators are in common use, conventional and automatic. The conventional type requires manual tuning to their dimensions to achieve suitable resonance, while automatic ultrasonic generators (Fig. 2), have an inbuilt system inside which automatically alters the output frequency to match the exact resonant frequency of the horn/tool assembly [21]. Nowadays, ultrasonic generators of frequency 20 kHz are more common in metal processing and welding than 40 kHz. More prominently, generators are designed to operate on a band of specific frequency (19 kHz to 21 kHz) rather than a single band (20 kHz). Digital ultra-



Fig. 1. Ultrasonic generator for 20 kHz output [23,24].

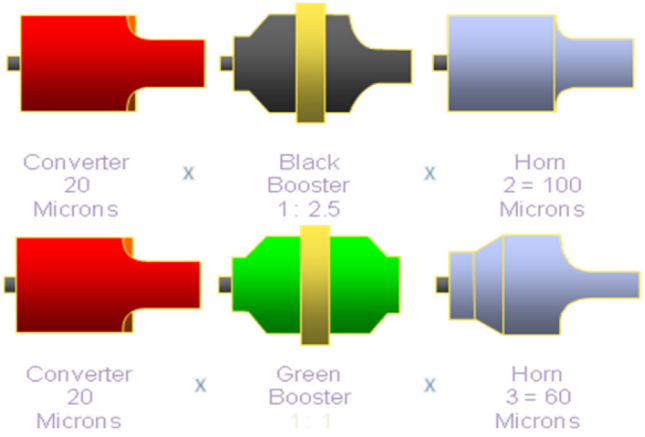


Fig. 2. Sequence of converter, booster and horn with their typical amplitudes [22].

sonic generators are designed and manufactured which facilitates the digital eyeshot of the frequency shift. Moreover these shifts can also be viewed to the computer screen, which facilitate import of real time frequency shift data during ultrasonic applications. This makes the applications exceedingly suitable of real time monitoring and control. Power variation system inbuilt in automatic generators, enables the users to work on different ranges of powers as well as amplitudes. Amplitude obtained from a 20 kHz converter is usually 12–20 μm [22], which is further amplified by booster and horn. A typical assembly of ultrasonic system showing the stacking sequence and amplified waveform at nodes and antinodes is shown in Fig. 2.

Automatic generators are also inbuilt with intelligent technologies such that they automatically adjust minor errors occurring due to wear and other irregularities, ensuring the energy loss to the minimum [25]. The power rating of the generators depend on the transducer specifications [26]. The automatic ultrasonic generators have also the safety features like auto switch off when some accident occur during malfunctioning or failure of horn or tool etc. [26–28].

2.2. Transducer

Ultrasonic transducer or converter is a devices which converts ultrasonic waves to the desired electrical signals or vice versa. These devices work on the principles similar to that used in radar and sonar systems. The converter is aptly named because it receives electrical energy from the power supply and converts it into mechanical vibrations [25,29,30]. They are designed to attain the vibrations either in longitudinal or compressive mode. Two types of transducers are most widely applied in industrial applications, magnetostrictive [26–28,31–35] and piezoelectric types [36–40]. The magnetostrictive transducers are capable of providing vibrations over a wide frequency range (17–23 kHz for a 20 kHz generator) which provides higher flexibility in the design of horn and wear adjustments [41]. Moreover the horn can be reused after its regrinding/machining [42–44]. Major issue with the magnetostrictive transducer is the electrical loss in the form of heat which adds extra cost and weight of cooling system [35,45,46]. A piezoelectric transducer consists of one or more disks of piezoelectric material sandwiched between two metal sections. The entire assembly is resonant at a predetermined frequency, and its length is acoustically equal to one-half wavelength at the applied frequency. It is placed just after the generator.

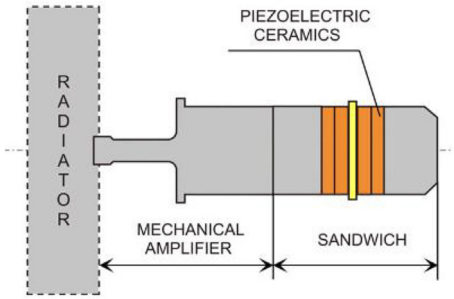


Fig. 3. Basic structure of the transducer [10].

Some piezoelectric transducers [21,46–49] also constitute a pair of synthetic ceramic plates made of lead zirconate titanate [36,50]. Piezoelectric transducers boast of high energy efficiency (90–97%), but still may require a cooling system [37,38,45]. These transducers are resistant to thermal failure and easy to construct [49]. Fig. 3 shows schematic of a transducer which comprises of a series of piezoelectrically activated vibrator plates stacked in the form of a sandwich. The whole system is placed in a metallic casing. Fig. 4 shows wide variety of transducer shapes used for industrial applications.

2.3. Ultrasonic horn

Ultrasonic horn is referred by different names such as acoustic coupler, velocity/mechanical transformer, tool holder, concentrator, stub or sonotrode. Owing to the negligible amplitude at the end face of transducer, ~0.001–0.1 mm [33,39,51], the horn is integrated into the ultrasonic setup so as to enhance the amplitude of ultrasonic vibrations [25,29,30,32,46]. Material of the horn is selected on the basis of the area of its application. For example, aluminium is ideal for normal applications while steel or titanium are appropriate for high pressure and complex applications. A horn material should possess good soldering and brazing characteristics, good acoustic properties and high fatigue resistance at high oscillation amplitude [51]. Besides, the horn material should be resistant to corrosion, heat, high wear resistance, have good elastic and fatigue strength properties, and high values of toughness



Fig. 4. Different types of transducers [24].

and hardness [29,52,53]. Some commonly used horn materials are titanium alloy, stainless steel, polycrystalline diamond (PCD), aluminium and Monel aluminium bronze [54–59]. The horn can be attached to the machining tool using different methods such as soldering, brazing, screw/taper fitting etc. Depending upon the requirement, horn can also be machined to give actual tool shape and size [26,43,52,53,56,59,60]. Threading makes the horn-tool attachment quick and easy, however, such joints are associated with issues of self-loosening, fatigue failure, and loss of acoustic power etc. [61]. The repetitive high frequency vibro-impact mode brings some unique properties and improvements into metal cutting process [62–66], where the interaction between the workpiece and the cutting tool is transformed into a micro vibro-impact process. Fig. 6 and 7 shows some commonly used horns shapes in metal processing and welding.

2.4. Booster

A booster, also known as amplifier, is an optional component in ultrasonic system, which is used when the amplitude at the end of horn requires some alteration. It is mounted between the converter and horn. The main function of booster is to amplify the resultant vibration amplitude of transducer. Generally, the diameter at the end attached to the horns is kept smaller to ensure that the booster increases the amplitude passing to the horn. Inversely, a greater diameter at the end attached to the horn would make the booster to decrease the amplitude. When clamped at their nodal point, booster horns provide rigid support for the converter/horn assembly. Typical configurations of different types of boosters are shown in Fig. 5.

2.5. Design of horns

Several theories and arts of designing horns have been proposed in the past. However, a unanimous understanding of the horn design is still lacking [68–73]. Primarily, two approaches are



Fig. 5. Boosters of different gain ratios and colour codes [67].

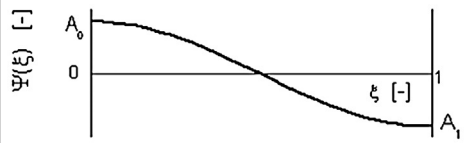
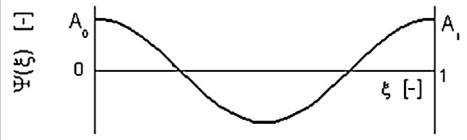
Name of Shape	Parameters	Mode Shape of Sonotrode Vibration
$\lambda/2$ – *"half wave"* shape	$k = 1$ $\lambda_1 = 2$ $f_{01} = \frac{1}{2l_0}\sqrt{\frac{E}{\rho}}$ $\vartheta_1 = 1$	
λ – *"wave"* shape	$k = 2$ $\lambda_2 = 1$ $f_{02} = \frac{1}{l_0}\sqrt{\frac{E}{\rho}}$ $\vartheta_2 = 1$	

Fig. 6. Mode shapes of cylindrical sonotrode vibrations [62].

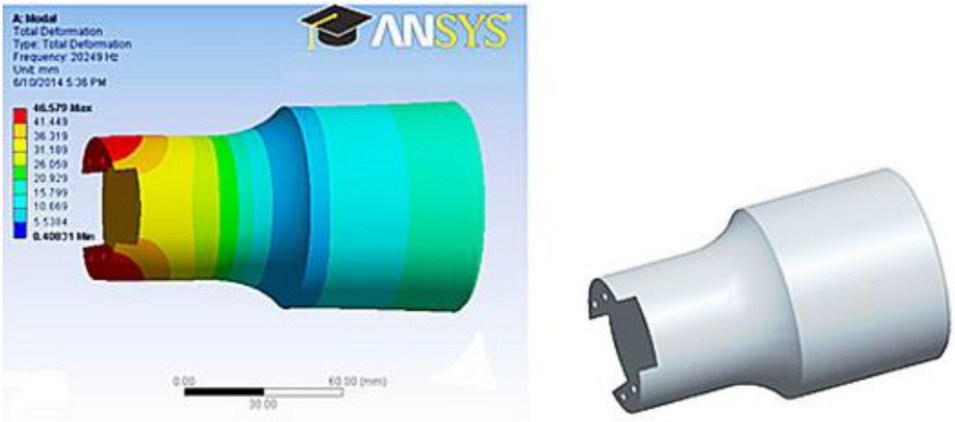


Fig. 7. Design of ultrasonic horn using FEM [80].

widely adopted in horn designing: traditional and finite element modelling (FEM). The traditional approach is based on application of differential equation where equilibrium of an infinitesimal element is taken into the consideration under the influence of elastic and inertia forces [73–75]. Length of the acoustic horn is a function of its working frequency and it does not play any role in oscillation amplification. The working frequency can be adjusted by tuning the horn. Tuning is an important part of horn design to achieve the resonance at the horn end and is usually performed at the transducer end where a tuning allowance of 10–15 mm is made [55,75–77]. Common horn designs in use are conical, exponential, stepped, circular and cylindrical types [75,78,79].

In the present era, finite element modelling (FEM) is the research trend of designing horn [73,78,80] because FEM is fast, user-friendly and yields better results. Moreover, complex shaped horns, which are very difficult to design by the traditional approach, can be easily designed using FEM approach. In the horn designing, FEM considers the weight of the tool, its material, boundary conditions and working frequency ranges. FEM can be effectively used to determine the working stresses on the horn in service to safeguard the horn from undesirable and sudden failures [70]. Nad [62] have investigated the designing of various shapes of horn to determine the optimum shape for ultrasonic working. Based on a systematic study on half wave and full wave mode shapes of sonotrode vibrations, as shown in Fig. 6, he suggested that different shapes of horn possesses different properties and can be used for separate purposes depending upon the requirements.

Fig. 7 shows the modal analysis of ultrasonic horn designed using FEM [80]. A horn can be designed and manufactured to achieve mixed lateral and longitudinal vibration modes [81]. The horn and the tool are designed to comply the FEM requirement to obtain the combined resonance frequency at a time. The horn design is an important aspect because even a minor excessive tool length can cause shift of resonance frequency and thus, variation of amplitude [70,82]. The recommended tool length to diameter ratio is of less than 20:1. For large size of tool, an equivalent amount of machining of horn is required to match the working frequency of both tool and the horn. Fig. 8 shows customized shapes of ultrasonic horns used in some complex processes.

3. Application of ultrasonic vibrations in welding

Ultrasonic vibrations have been applied in various welding processes either to weld metals directly by ultrasonic action, as in ultrasonic welding, or to assist the metal process for improving the process effectiveness, feasibility and product quality, as in various fusion welding and solid-state welding processes as enumerated below.

3.1. Ultrasonic welding

Ultrasonic welding is a process of joining two parts by using ultrasonic vibrations of high frequency that convert electrical energy into heat. It is a suitable process to join plastics as well as most of the metals. In this process, high frequency vibrations (in the range of 19–22 kHz) in combination of pressure are applied to join two materials quickly and securely, without the need of large amounts of heat. Being a non-consumable technology, ultrasonic welding can be employed to weld dissimilar metals with large difference in their melting points. Fig. 9 shows a typical ultrasonic welding system. In the system, two types of forces are in action, a static normal force applied perpendicular to the interface between



Fig. 8. Customized horns designs used for complex applications [83].

the workpieces and an oscillating force generated by oscillation of the contacting sonotrode parallel to the material interface surface [84].

The relative motion at the beginning of the welding cycle cleans the surfaces in contact with each other and plastically deforms asperities [85,86]. The combined effect of static and oscillating forces produces deformation which promotes welding [86]. In the most widely used versions of the wedge-reed system, the anvil is a self-vibrating element, which flexes similar as the reed. In particular, the anvil is designed to be contra-resonant [87], and vibrate in a manner to achieve an increased relative motion at the workpieces. Hence, in the wedge-reed system, electrical power is converted into high frequency mechanical longitudinal vibrations by employing a transducer. Subsequently, the longitudinal vibrations are amplified and transmitted into the reed by a metal wedge, brazed to the reed, and converted into transverse vibrations.

Recently a new technology named as Meta-Lax Weld Conditioning (MLWC), further renamed as Pulse Puddle Arc Welding (PPAW) came into existence to address problem of weld distortion and cracking. Further these technologies are economical, convenient, and effective in preventing even other welding issues also which warrants a rapid integration in the ultrasonic welding industry.

3.2. Ultrasonic vibrations in metal welding and spot welding

Ultrasonic metal welding (USMW), invented in Germany during the second world war, is a solid state process in which two parts are welded by the application of ultrasonic vibrations under moderate pressure. Little or no heat is applied in the process and the process temperature reaches ~50% of the melting temperature material. Like other solid state welding, the joint formation in USMW is a three stage process [88–96], as presented in Fig. 10. In the first stage, the upper specimen vibrates with the welding tip while the lower specimen remains stationary. Hence, a substantial relative motion

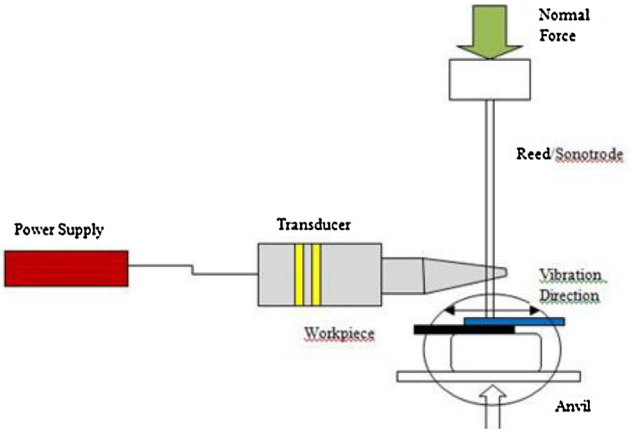


Fig. 9. Principle of ultrasonic welding set up for spot welding.

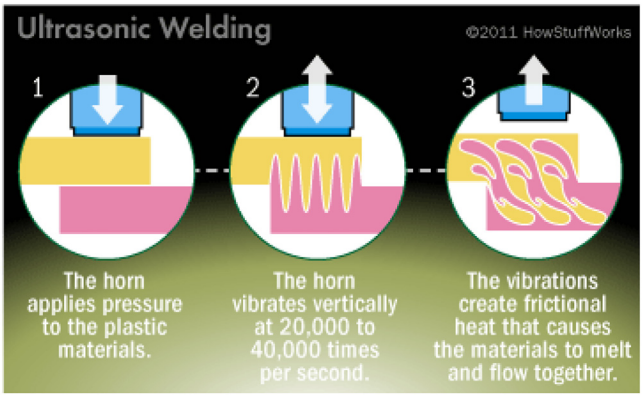


Fig. 10. Stage of ultrasonic welding [100].

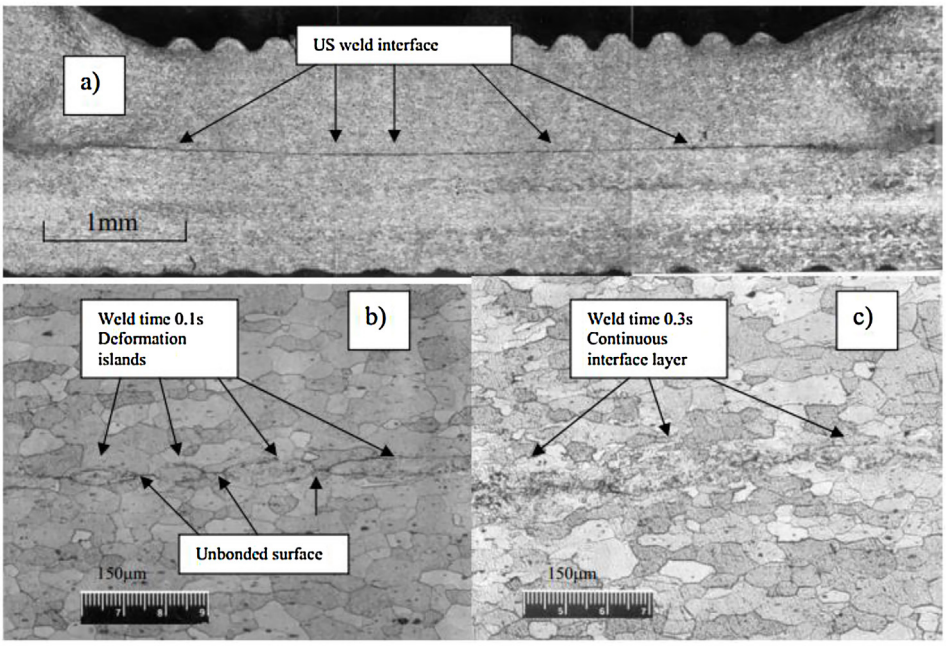


Fig. 11. ultrasonic weld cross section, (a), deformation islands after 0.1 s (b), and high magnification picture of the continuous weld layer (c) microstructure determination [102].

takes place in between the upper and lower specimens. During second stage, the relative motion is supressed in the partially bonded region, however a relative motion persists between the weld tip and the upper specimen. In the third stage, the two metal plates plastically deform with rise in temperature. The partially bonded region grows in size and the joint strength reaches to a maximum value. Materials suitable for ultrasonic welding include nonferrous metals and their alloys, however, ultrasonic welding of some materials such as zinc (Zn), lead (Pb), and tin (Sn) alloys is yet to achieve desired success.

Wodara [97] has summarized the three stages of bond formation [93–96] by explaining different bonding layers during the ultrasonic welding. First, a longitudinal mode ultrasonic vibrations were applied to the workpieces in conventional resistance welding equipment with an objective of lowering the surface resistance in spot welding of aluminium (Al) [98]. During this experiment, it was observed that ultrasonic vibrations alone are capable of producing metallic bonding. Since the mechanism of metal joining in USMW is adhesion or bonding at the interface layer of the two faying surfaces [99–101], it is immune to the thickness of the material [101]. Intense material deformation at the plastic interface produces bonding between the metals.

Fig. 11 shows typical microstructure of an ultrasonically welded aluminium [102]. It can be observed that the interface represents a very thin layer of the deformed material. Due to Ultrasonic vibrations, the layer formed has very fine grain structure which results due to heavy plastic deformation. It also shows few zones of discrete deformation of the welded sections. The welds formed with minimal or less external deformations presented higher strength [102]. It has been observed that microwelds started to form in the vicinity of contact surfaces as soon as the heat is imparted [103,104]. These microwelds spread in the zones and combine with adjacent microwelds to generate larger welds [101]. Some authors have considered the sliding during welding cycle as fretting action between the two surfaces for a full welding cycle [86,105]. Rozenberg have observed that the deformation zones were oriented parallel to the direction of impingement of ultrasonic vibration [50]. It has been found that sliding between the sonotrode and anvil causes their mutual sticking [102] which creates hurdles in the ultrasonic welding. Therefore, efforts are made to minimize sliding between the sonotrode and anvil. Tsujino et al. [106–109] have conducted a series of investigations and proposed a new method to weld metals and plastic materials by using the effect of ultrasonic waves on different sizes of metal specimen and plastic. In their investigations, ultrasonic butt welding was successfully used to perform end to end joining of thick metal sections. Further, they investigated the effectiveness of two vibration systems crossed at right angles. The complex type vibration system was found to be effective for end to end lap or butt welding of thick and medium sized large metal parts and for plastic welding at frequency above 20 kHz. Further they [107] studied the dissimilar butt welding of aluminium alloy to stainless steel by using ultrasonic vibrations of frequency 15–25 kHz. They observed that the hardness of aluminium side dropped down the welded surface while that of at stainless steel side dropped in the zone 2 mm away while, it increased by ~20% in the zone adhering the welded surface. The improvement in hardness has been attributed to the partial hardening of material due to the ultrasonic vibration while the reduction in hardness has been associated to relaxation of residual stress during the vibratory effect at the surface. Tsujino et al. have also shown that successful end to end ultrasonic butt welding of aluminium alloy, anticorrosive aluminium alloy and copper is feasible under the 19 kHz ultrasonic vibration welding system [108]. However, thin burrs generated in upper and lower sides of the welded region and relatively low hardness of the zone adjacent to the weld surface are the major disadvantages. The lowering in hardness was legible for pure aluminium alloy while it was sizable for the anticorrosive aluminium alloy. Matsuoka et al. [110] investigated ultrasonic welding of dissimilar metals such as aluminium-copper in the air and under water. Their experiments showed that the values of welding pressure and welding time for underwater welding were higher than those for atmospheric welding. However, the strength of weld metal in both the cases were similar. In a comparative study on ultrasonic welding of aluminium using low and high frequency vibrations, neither diffusion nor recrystallization was found to play a role in USMW [111,112]. This was attributed to the insignifi-

cant effect of vibration frequencies on process temperature. The above study reported a layer of deformed material of approximately 30 μm similar to that reported by de Vries [102]. Heymann et al. [113] investigated the effect of specimen surface on process effectiveness in ultrasonic welding. They found that sonotrode adhesion and weld quality increases with sample cleanliness. Microstructure of the ultrasonically welded specimen revealed that the grains size in a very thin weld layer is remarkably small [97,114], however, the ultrafine grains did not correlate well with the thermal history of the process. Boothroyd and Knight, studying the diffusion in ultrasonic welds, showed that diffusion occurs along the grain boundaries rather than the bulk of the grain in a weld [45]. It was also affirmed that metallic adhesion is the primary mechanism of bond formation in USMW. Watanabe et al. [115] studied the effect of ultrasonic vibration on weld metal of ferritic stainless steel. They observed that ultrasonic vibrations facilitate formation of equiaxed grains and their accumulation in the middle of the weld. Tensile strength and elongation of welded section were considerably improved with ultrasonic vibration as compared to those without ultrasonic vibration. Javadi [116] suggested that minimum power required for optimum clamping depends upon sheet thickness and type of metal welded. His study showed that producing a good weld is a function of input power, welding time, work piece dimensions and welding material. Joshi et al. [117] studied the feasibility of metal–metal ultrasonic welding and the bonding between the metals. Their study inferred that a good bond formation in the ultrasonic welding of pure face centred cubic metals (Al, Cu, and Au) depends upon optimum welding parameters albeit without any interfacial sliding and non-optimum heating. Jeng and Horng [118] used an ultrasonic bonding machine and a pull tester to analyze the effect of applied pressure, surface roughness and input power on ultrasonic bonding. They showed that the lower value of applied load or input power makes the welding of wires and pads feasible. Chang and Frisch [103] developed the concept of the bonding system and explained the development of bonded area due to the ultrasonic vibration. They summarized that bonding is accomplished via two different processes: interfacial slip and sub layer plastic deformation along with a localized temperature rise at the weld. The maximum average interfacial temperature produced in the weld zone ranged between 35–50% of the absolute melting temperature of the material. Ultrasonic vibrations were also used to join metals with glass/ceramics, As glass/ceramics undergo very little deformation, the final joint totally relies on the strength of their interface layer formed by diffusion of the two materials involved in the joining [119,120]. For example, if aluminium is joined to glass, the interfacial layer would be aluminium oxide. During USMW, temperature plays an important role for the formation of good and defect free joints. Therefore, investigation of heat distribution and temperature rise in the vicinity of welded joints is important. A theoretical study on heat dissipation in the ultrasonic welding of aluminium alloy reveals that the approximate temperature in the weld zone is 600 °C [101], which is 60–80% of the melting temperature of aluminium [121,122]. It shows that temperature plays an important part in weld formation during ultrasonic welding. Richter [123] experimentally affirmed the above suggestions. Further, the temperature rise of weld zone is depends upon the amount of the energy transferred by ultrasonic vibrations. The energy supplied can be measured by several methods such as electrical power measurement, measurement of standing wave ratio by strain gauge, measurement of acoustic power, use of calorimetric methods and thermocouples [124–127].

Ultrasonic spot welding (USW) also have a very long history and mainly invented to cater the industrial demand (especially in aerospace and automobile) of substantial weight reduction with increase in strength. Usage of thin sheets of aluminium in the automobile and aerospace application become very popular due to its several advantages such as high strength-to-weight ratio, environment friendliness, good formability, machinability, and recyclability [128–130]. However joining of thin sheets of Al to Al and other metals is very difficult by fusion welding processes owing to the reason that they melts very quickly with small heat. Resistance welding (RSW) process also used to join the thin sheets however the weld quality is not up to the craved level. Moreover it was ascertained that the weld was suffered from several defects leading to poor mechanical and microstructural properties. High conductivity, low strength at temperature, tendency to degrade the electrodes and high energy cost makes it unsuitable to join Al alloys [131,132]. Past studies have also evaluated that in RSW, a high rate of intermetallic reaction takes place in the dissimilar metal joining, which severely affects the weld quality [133,134]. Alternative fusion welding option were investigated such as clinching, adhesive bonding, laser welding and GTAW, or GMAW [135,136], however they are appeared to be endured from several issues such as poor weldability, high distortion, severe defects, poor mechanical properties, non-uniform microstructure, addition of consumables, expensive, high energy demands, and additional surface treatment cost [135]. While moving to solid state welding, friction stir spot welding (FSSW), is seemed to overcome some of its issues however the weld cycle is found to be quiet long (2–5 s) [136,137].

Another solid state joining process received a great attention of the researchers named as USW [132,138–140] was invented that could potentially replace several shortcomings occurring in fusion welding processes. Studies affirmed that USW claimed several advantages as compared to fusion welding processes in terms of good mechanical and microstructural properties, low energy requirement (using ~2% of the energy of RSW), shorter weld time, no shrinkage and distortion problems, higher weld efficiency, less or no defects, and no HAZ damage [132,139,141,142]. USW technology proved to be exceedingly efficient while joining sheets in case of mass production making it very economical than other fusion joining processes. USW is found to be more efficient than FSSW because the energy is predominantly produced at the center line of weld [143] rather than at the top surface. Several Industries also adopted USW for spot welding the frames and structures of aluminium and other metals because of its intended results [132,138].

In a very short time, Chen et al. has accounted legion authoritative outcomes in the field of USW of dissimilar metals such as Al to Cu, Al to Mg, Al to Ti, Al to steel as well as that of composites [128,144–148]. They also studied about the microstructure, texture and fatigue properties of these joints and concluded their last longing findings. Macwan et al. investigated the USW of Al to galvanized high-strength low-alloy (HSLA) steel [144]. They narrated that the value of tensile lap shear load was seemed to be increased with increase in welding energy which reached to a highest value, and then further decreased with a further rise in energy as shown in Fig. 12. In a similar attempt of joining Al to steel, Haddadi et al. concluded that due to presence of thick Fe_2Al_5 and $FeAl_3$ phases, the interface between them lowers the lap shear strength and it reaches to 1.7 kN form the 3.1 kN which is also found to be a function of temperature [149].

Macwan at al also claimed that the USW samples welded at 2000J shows higher fatigue life as compared to those welded at 1000J [144]. They also noticed a tiny micro-level crack tip at the notch of two sheets that produced a stress concentration effect during cyclic loading. It admitted the cracks to develop in the transverse direction during USW of Mg–Mg joint [150] and RSW of DP600 dual phase steel [151]. Similar investigations are performed by Patel et al. in their study [128]. Prangnell et al. successfully joined 1.0 mm 6111 aluminium and DC04 steel [98]. They affirmed 2.8 kN of maximum failure load in lap shear test that was 95% of that in Al–Al welds. Balasundaram et al. studied about USW of Al to Cu joint while using Zinc interlayer [145]. They claimed that micro-

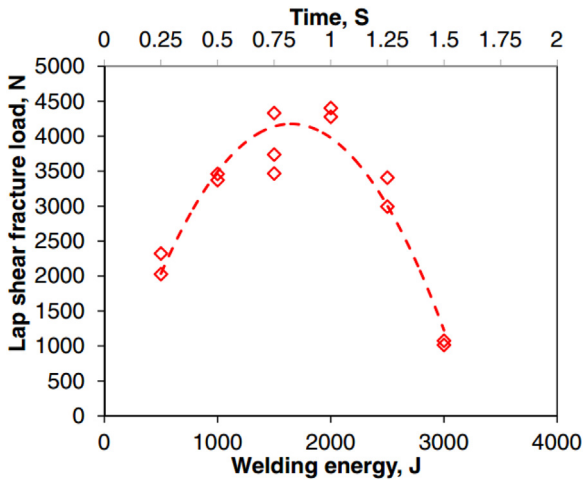


Fig. 12. Lap shear load as a function of welding energy or welding time at a constant power of 2000 W and clamping pressure of 0.4 MPa [144].

hardness of USWed joint was decreased with increasing energy input which can be attributed due to appreciable grain growth at high temperature which supports their claims made during early studies [152,153]. Besides, Zn interlayered joint is seemed to have larger lap shear tensile strengths and fracture energy than that of without Zn interlayered joint. Yang al [86] performed resistance heat assisted USW of Al and Cu. They found out that peak power as well as the peak temperature of the weld interface were increased drastically. The weld resulted good metallurgical binding of Al and Cu. The lap shear load was found to have a maximum value of 550 N at the DC level of 1100 A. Zhao et al. [154] investigated the welding energy during USW of Al-Cu. They narrated that joint strength was seemed to be increased with the welding energy and reached the maximum 3.3 kN at 1000 J. They observed a swirl type shape at weld interface which leads to a mechanical locking in between the metals and improves the strength [154].

Nowadays ultrasonic metal welding is being integrated with the complex vibrations including bending vibrations which were proved to be more efficient than axial vibrations. The technology is being further advanced using advanced simulation software which predicts the optimum designs resulting minimum ultrasonic losses.

3.3. Ultrasonic vibrations in arc welding

Success of a welding process in the industrial sector relies on high production rate at low cost and good mechanical and metallurgical properties of the product. Conventional arc welding processes can be accomplished with low operation cost due to the application of voltaic arc which is of low cost, easy to control and portable. However, conventional arc welds are vulnerable to several types of defect and improper grain refinements. During arc welding processes, the grain structure of weld metal is coarsened thereby leading to deterioration of weld mechanical properties [155]. The above issues continue to be partially challenging despite several interesting developments. In a series of studies, application of mechanical vibrations into arc zone was found to be beneficial in improving the structure and properties of welded joints [156–160]. Integrating the mechanical vibrations with the arc process is an important aspect of the above development. Several techniques can be adopted to impart the mechanical vibrations in the welding zone, such as electric-arc vibrations [161,162], thermal effects of a coherent laser beam [163], direct introduction of the vibrations into the material through a waveguide [164], electromagnetic stirring [165,166], introduction of ultrasonic vibrations [166–168], weld pool oscillation [169], oscillation of welding arc [170], vibration of electrode [171], molten droplet oscillation [172], and vibration of workpiece during welding [156]. These techniques require an additional energy in the weld pool to promote and enhance the nucleation as well as grain refinement. The welding arc is integrated with the ultrasonic vibration setup so that an ultrasonic induced arc oscillating at the ultrasonic frequency can be generated [161,173,174]. The ultrasonic vibration assisted arc welding is found to be more promising in terms of improving the weld mechanical properties and microstructure [175,176]. Dong et al. [156] used mechanical vibrations of different frequencies to weld high-strength aluminium alloy. They observed higher grain refinement, stable weld structure and reduced probability of hot cracking. The reduction in hot cracking as well as the melioration of weld structure and properties were confirmed from experiments using the same frequency mechanical vibrations [175,177–182].

3.3.1. Ultrasonic vibrations in GTAW

Gas tungsten arc welding (GTAW), also known as tungsten inert gas (TIG) welding is an arc welding process that uses a non-consumable tungsten electrode to form the welds. Inert gases such as argon or helium are supplied into the weld zone in the form of shielding gases to protect the weld area from atmosphere contaminations. GTAW proved to be a low heat input process [183]. It can be easily integrated with controlled automatic wire feeding system which is mainly adopted in robotic welding for mass production. Ultrasonic waves were recently integrated with GTAW process with the aim to improve the penetration depth by oscillating the torch [170]. It was also seen that when ultrasonic waves were employed, the material mixing in the weld pool in bead-on-plate weld was enhanced [177]. Using an ultrasonic integrated GTAW process, efforts were made to visualize the effect of vibrating arc on the porosity and solidification structure of commercially pure aluminium [168]. Thus, monitoring and control of automated welding machines has been an active research area in the past few decades [160,184–192]. Various techniques used to monitor and control the welding operations are pool oscillation based control [184], radiography based control [185], thermal based control [186,187,192], and vision based control [188,192] etc.

Sun et al. [170,178,194] introduced ultrasonic vibrations into the GTAW process by employing a piezoelectric transducer made of titanium wire I ntegrated in axial direction with the electrode (Fig. 13). The ultrasonic vibrations were applied to the electrode axially and were transmitted to the welding pool. It was seen that ultrasonic vibrations improved the weld pentation depth and width of AISI 304 stainless steel weld. Similar results were also shown by the Fan et al. [195]. In both the above studies, ultrasonic application improved the weld quality [170,177]. A comparative study of conventional GTAW and ultrasonic vibration assisted GTAW under identical experimental conditions showed that at welding currents,100 A and 150 A, presence of ultrasonic vibrations caused the increase in width ratio by 235%, and 380%, respectively while the weld pentation depth was also increased [178]. The above improvements were attributed to a deeper weld pool penetration imposed by ultrasonic vibrations which enhanced the weld pressure and caused increase of weldment. It was also found that ultrasonic vibrations improved the arc pressure which was not found in the case of conventional GTAW process [196]. Application of ultrasonic vibrations on the welding arc in GTAW caused arc constriction and increased the weld penetration depth [193], as shown in Fig. 14. Krajewski et al. [157] studied the effect of ultrasonic vibrations on aluminium alloy. They showed that weld quality can be enhanced only when the substitutive stresses (σ_R) are compressive in nature at the faze 3/4λ nodes. Grains of the weldment were refined and weld porosity was reduced. The integration of ultrasonic vibrations caused significant changes to the bead structure at a distance of 0.127 m from the waveguide front end, as shown in Fig. 15. Also,

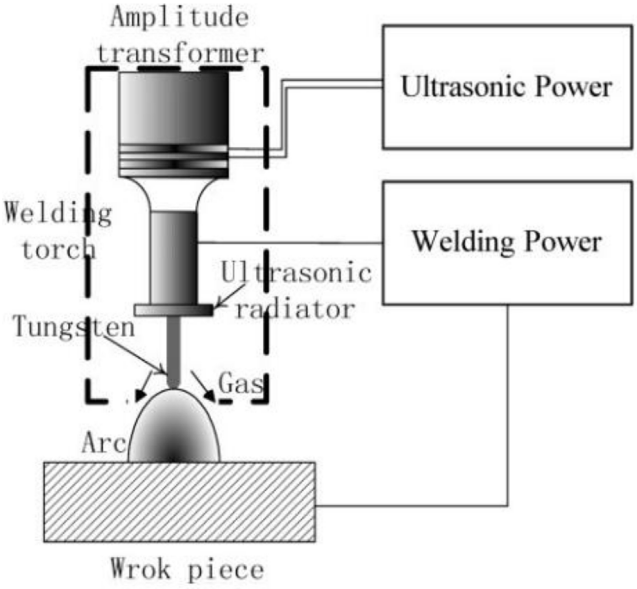


Fig. 13. Schematic of ultrasonic assisted TIG welding system [170].

the face width and the weld penetration depth were substantially increased during the above ultrasonic assisted GTAW process. Dai [164] also integrated ultrasonic vibrations perpendicular to the face of workpiece in GTAW. This study reported that the presence of ultrasonic vibrations induced a ~45% increase in weld penetration depth and a ~42% reduction in grain size as compared to conventional GTAW. Al–Mg alloy with galvanized steel and Al–Mg alloy with stainless steel. Joining of both the above combinations were reported to be successful. Also, substantial grain refinement and the associated improvement of tensile strength was observed in the Al–Mg alloy-galvanized steel dissimilar weld.

Further, Dai [164] employed ultrasonic waves to perform dissimilar lap welding of To integrate with the GTAW process, ultrasonic vibrations were also applied in the direction parallel to the displacement of welding torch [167,174,197]. In this configuration, analysis of weld joints prepared with different welding speeds showed improvement in weld tensile strength when compared with the conventional GTAW process. It was observed that weld tensile strength as well as the width of austenitic grains increased with increasing amplitude of ultrasonic vibrations [160,167]. Watanabe et al. [115] induced ultrasonic waves into the weld pool through a wire filler metal with the guide tubes attached at the end of ultrasonic sonotrode. They reported that the grain

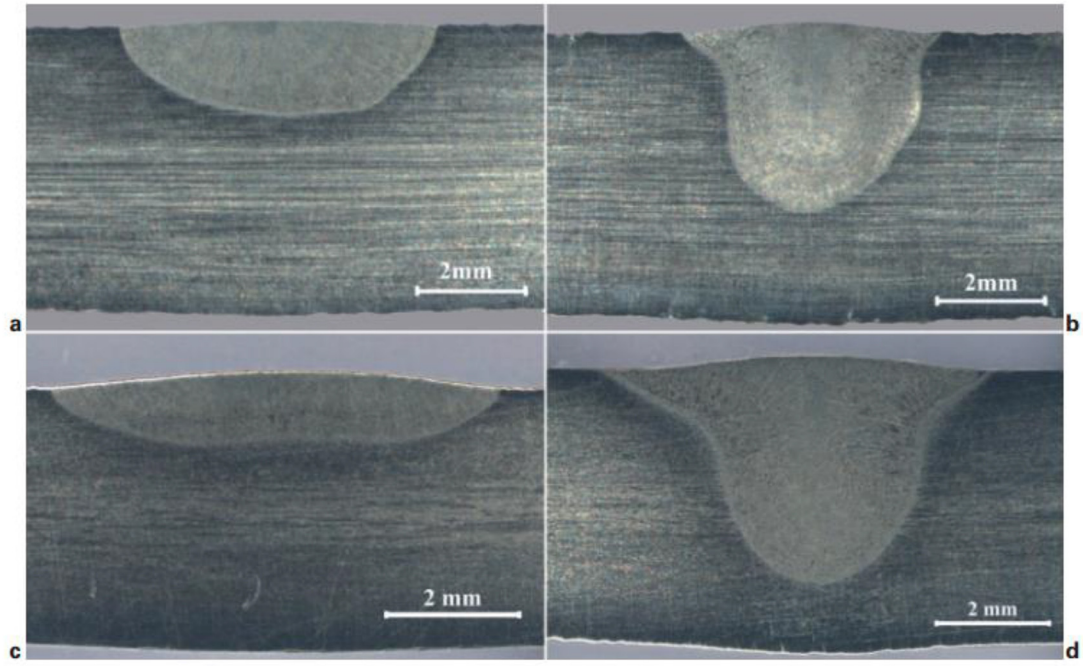


Fig. 14. Macrographs of weld on cross-section samples of AISI 304 joint (a) 100 A without ultrasonic; b 100 A with ultrasonic; c 150 A without ultrasonic; d 150 A with ultrasonic [193].

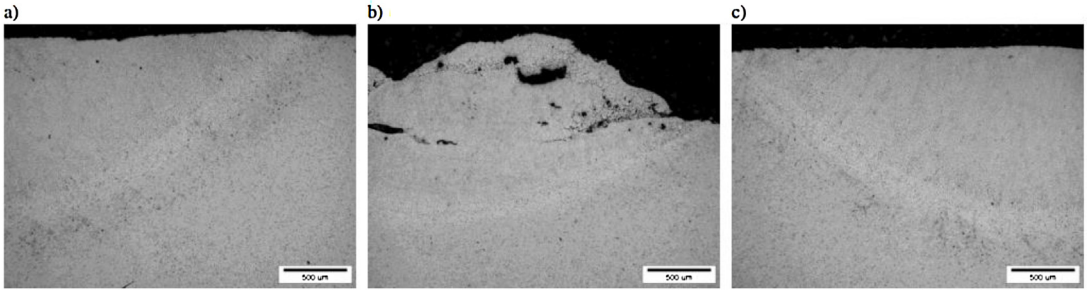


Fig. 15. Micrograph of the TIG DC welds: (a) without ultrasounds, (b) with the participation of ultrasounds z = 0.127 m (1/2λ), and (c) with the participation of ultrasounds z = 0.1905 m (3/4λ) [157].

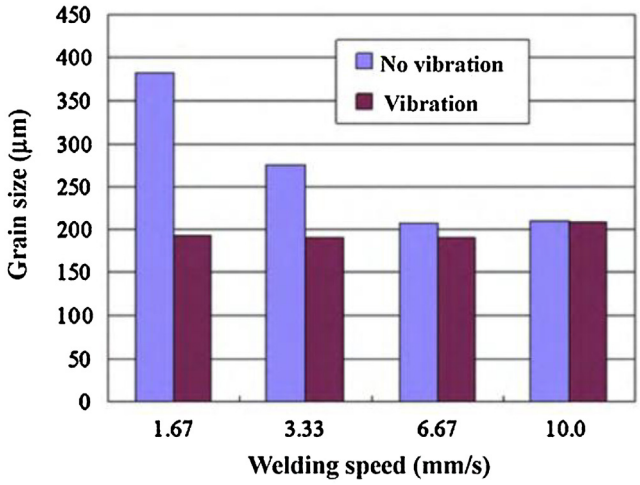


Fig. 16. Average grain size of the weld metal made without and with ultrasonic vibration using various welding speeds [115].

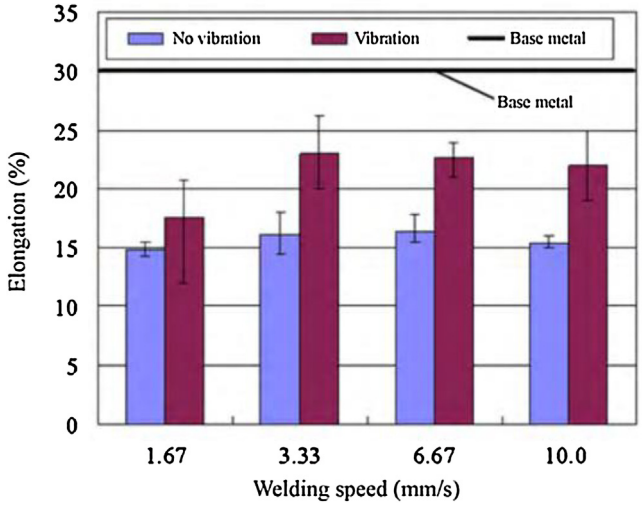


Fig. 17. Fracture elongation vs. welding speed when ultrasonic vibration was applied or not [115].

refinement and fracture elongation were considerably improved at low welding speeds, as shown in Figs. 16 and 17, while higher welding speed resulted in the formation of equiaxed grains at the centre of weld bead, leading to improved ductility. It can be seen that in the absence of ultrasonic vibrations, the crack is in the columnar grains grew perpendicular to welding direction while the presence of ultrasonic vibrations makes the crack propagate through the equiaxed crystals (Fig. 18). As the consistency of equiaxed grain growth in the tensile direction is low, the tensile strength of the weldment increases in the presence of ultrasonic vibration.

State of art in GTAW includes the development of Orbital GTAW as well as UaOGTAW (ultrasonic assisted orbital GTAW) technologies for developing high quality pipe joints. Further Ultrasonic assisted GMAW has found special attention in jewellery industry to melt, and join platinum group metals.

3.3.2. *Ultrasonic vibrations in GMAW*

Gas metal arc welding (GMAW) uses an electric arc, produced between a consumable wire electrode and workpiece, to melt and join the workpiece. Owing to its high productivity and low heat input, GMAW is widely used in industries for the welding of a varieties of ferrous and non-ferrous metals [156]. Production of good quality and defect free GMAW welds requires control of process parameters such as arc length, voltage and current ratings, wire feed rate etc. However, GMAW process is not a stable technique because it generates spatters and welding fumes [160,198–200]. The window of process parameters to achieve a stable arc is narrow because the welding voltage is very small, of the order of 2 V [201]. To address these problems, ultrasonic vibrations were integrated with the GMAW process in different fashions. The idea of applying ultrasonic vibration in GMAW was to bring about variations in metal transfer in GMAW by the ultrasonic effect and to utilize the ultrasonic induced grain refinement criteria which is expected to improve weld mechanical properties. It was seen that the GMAW weld structure and grain refinement increased considerably with ultrasonic vibrations [202–204]. Moreover the welding arc can also be integrated with the pulse current to generate the ultrasonic wave in the ultrasonic frequency range [160,166,167,175]. Thus, the arc source acts as the heat source as well as the ultrasonic wave emitter. Using this setup, extensive grain refinement and improved mechanical properties are realized. Fan et al. studied short circuiting transfer mode of ultrasonic assisted (UGMAW) method [205]. They found that the voltage range and arc stability was improved by using the UGMAW. It was seen that when ultrasonic vibrations are applied directly on the workpiece, a substantial grain refinement of weld occurs [175,180]. The schematic of UGMAW process used in their study is shown in Figs. 19 and 20. In case of UGMAW, the

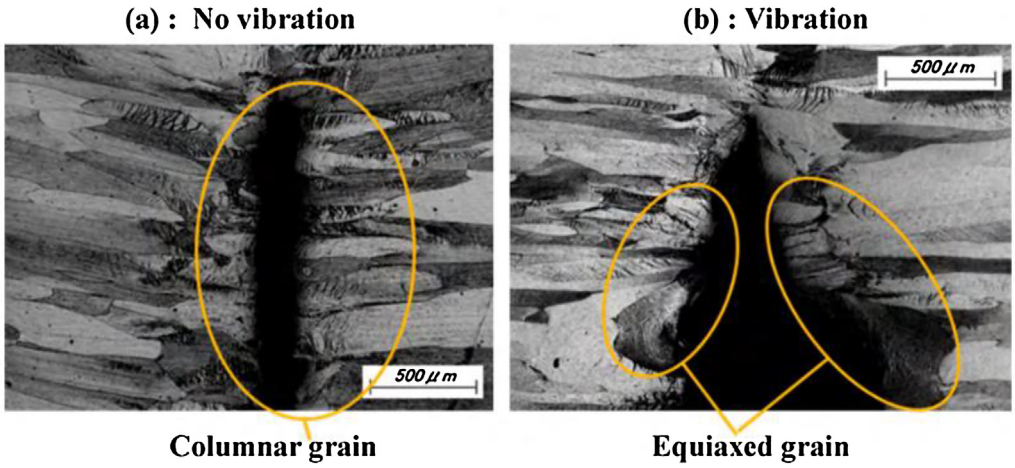


Fig. 18. Enlarged images of fracture paths in the weld metal made without and with ultrasonic vibration at the welding speed of 6.67 mm/s. (a) without vibration (b) with vibration [115].

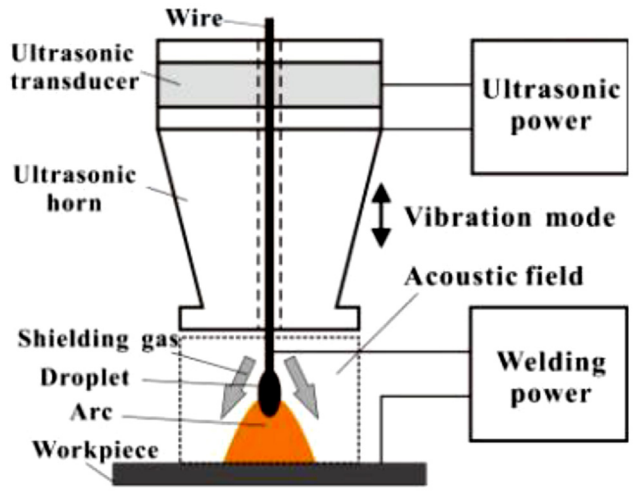


Fig. 19. ultrasonic assisted GMAW process set up [172].

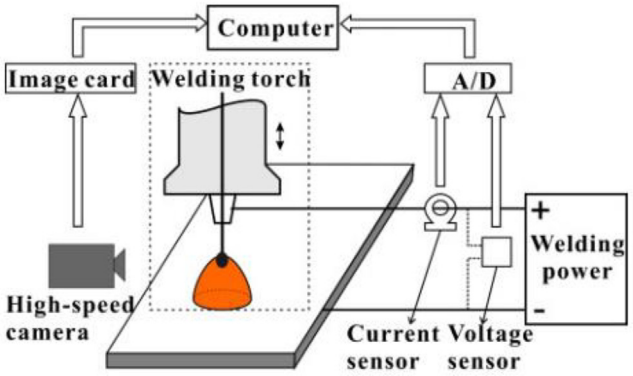


Fig. 20. transfer mode in the U-GMAW process are presented in comparison with the conventional GMAW process [206].

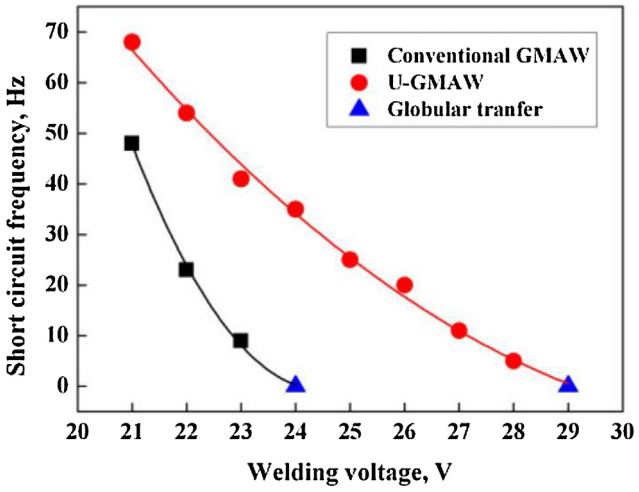


Fig. 21. Short circuit rates at different welding voltages [205].

short circuit frequencies are higher at identical welding parameters (Fig. 21).

It indicates that droplets transferred to the weld pool were relatively smaller in size which led to higher stability and better weld quality [198,201]. Some other authors also studied the effects of ultrasonic frequency on the geometry of drops [173]. They observed that at ultrasonic frequency 20 kHz, power 110 W and vibration amplitude 30 μm, the droplets begin to deform, the diameter of droplets becomes higher than the wire diameter and the droplet stretch along the wire axis.

This deformation process continues until the end of the droplet transfer cycle. During experimentation, they also noted an ellipsoidal shape of the droplet with its major axis along the wire axis and its minor axis measuring twice as the wire diameter. Further, they integrated ultrasonic vibrations in GMAW to be able to detach the droplets. It was found that the droplets became elongated along the wire axis but their size was reduced as compared to those in conventional GMAW. In a further study, Fan et al. [206] studied the influences of ultrasonic vibration amplitude, working area of the ultrasonic radiator and its distance from workpiece on the welding arc. Their study inferred that the arc in UGMAW was contracted, brightened and was of shorter length. The arc length was found to be a function of the height of the radiator and it increased with the radiator diameter. They derived that the amplitude of the stationary ultrasonic wave is a function of incident wave and reflected wave. A frequency of ~20 kHz and maximum amplitude of 45 μm yielded the most constricted arc whose length is smaller than the conventional GMAW [207]. It depicted that application of the ultrasonic vibrations caused increase in wettability [205,208]. Xie et al. studied the effect of acoustic field parameters on arc acoustic binding during ultrasonic wave-assisted arc welding [209]. They established a model for a cylindrical parameter where incident acoustic field was described based on boundary element equation. From the model, the maximum acoustic radiation power was found to be in first resonant mode. When applying the ultrasonic vibrations, the shortening of welding arc generated arc acoustic binding, as shown in Fig. 22 [210]. Further, the weld penetration was improved and grains were more refined, as shown in Fig. 23 [211].

State of art in GMAW includes integration of robotic application by improving the stability of the arc at low values of welding current. The previous studies were limited to aluminium, mild steel but new welding equipment has included structural steel and titanium. The increase of the welding current and the load gives additional advantages. It is also evident that enhanced weldability and process efficiency can be incurred for thick and hard metal sheets through advanced power source regulation, especially over short circuiting, controlled polarity and electrode wire motion.

3.4. Ultrasonic vibration in friction stir welding

Invented in 1991, friction stir welding (FSW) is more younger a solid state technology to gain worldwide acceptance [212–215]. Based on the large quantum of FSW research over the years, the fundamental principles, process mechanisms, influence of process parameters on process variables and weld physical and mechanical properties, the weld microstructure etc. are relatively well understood. Modelling studies of various aspects of the FSW process have also tasted their fair share of success [14,216–224]. Owing to its attractive propositions in the welding of Al to Al and Al to M (M: Mg, Cu, Fe, Ti and Ni) alloys which subsequently amount to considerable weight reduction and energy saving, FSW has been widely adopted for commercial applications in aerospace, maritime, railway, automotive and electronic sectors. At present, research efforts to employ FSW to join high strength and high melting point metals such as steel, Ti and Ni alloys are at full acceleration. However, conventional FSW is far from completely replacing the fusion welding techniques in terms of energy consumption, productivity and cost of production and feasibility due to its limitations such as heavy load requirements, low welding speed and tool wear. Padhy et al. have pointed out that several recent FSW variants that apply an external thermal or mechanical field have been developed in order to resolve the above issues [224]. Objective of the assistant energy fields is to reinforce material plasticization either directly or by

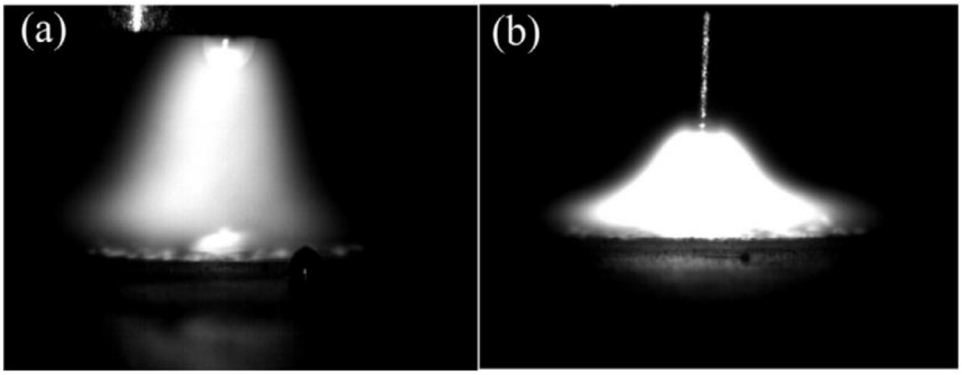


Fig. 22. The arc shape of (a) conventional arc and (b) ultrasonic wave-assisted arc [210].

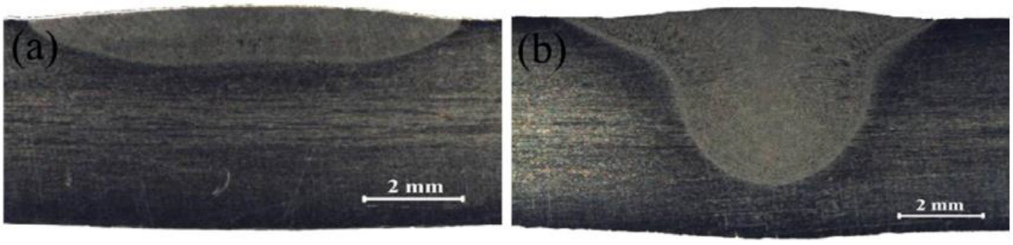


Fig. 23. Weld penetration of (a) conventional arc welding and (b) ultrasonic wave-assisted arc welding [211].

generating additional heat in the FSW process. Ultrasonic vibration assisted FSW (UaFSW) process is a very recent technology. Integration of the ultrasonic system to FSW machine is easy. The modes of ultrasonic exertion on FSW workpiece include exertion of the vibrations along the horizontal direction to FSW tool [80,225–228], along the axial direction to FSW tool [229–232], and to the workpiece ahead of the FSW tool [233–245]. Based on the modes of ultrasonic exertion, several types of UaFSW processes have come into limelight during the past 5 years [80,225,233–239,246].

3.4.1. *Ultrasonic vibrations exerted in horizontal direction to FSW tool*

In such a system, the ultrasonic vibration set up is integrated in the FSW machine such that the ultrasonic vibrations can be applied in horizontal direction to FSW tool as shown in Fig. 34.

The credit of inventing such system goes to Park et al. [226,227,246], who applied ultrasonic vibrations on the FSW tool in horizontal direction via bearings during the FSW process and studied the effect of vibrations on the process at different process parameters. They could achieve substantial reduction in axial force (F_z) which was clearly evident since the plunge stage.

They could reduce the axial force further by increasing the ultrasonic vibration amplitude. The trends of variation was similar for

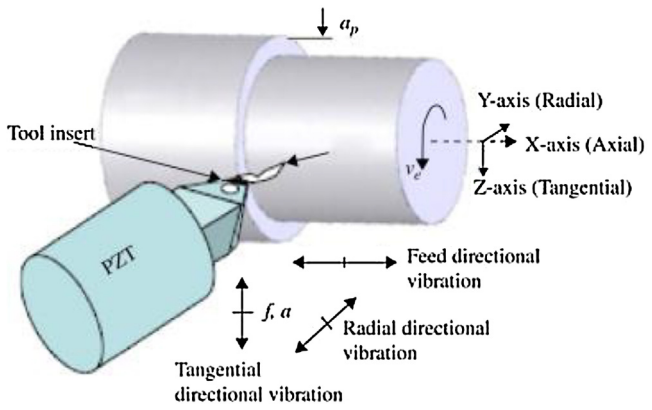


Fig. 24. Schematic of ultrasonic vibration cutting [274].

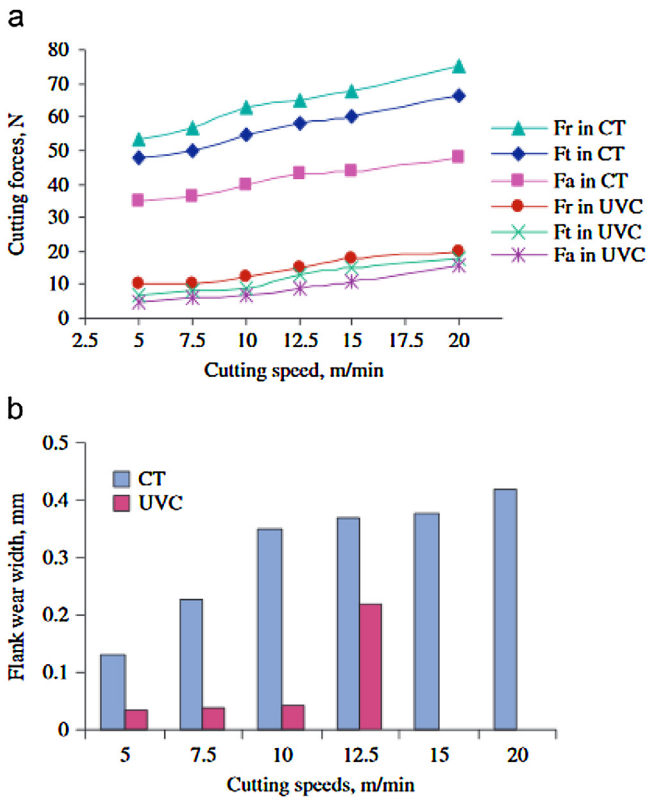


Fig. 25. Effect of cutting speed in cutting operations with and without ultrasonic vibrations: (a) cutting force components, and (b) tool flank wear width, after 10 min of cutting [274].

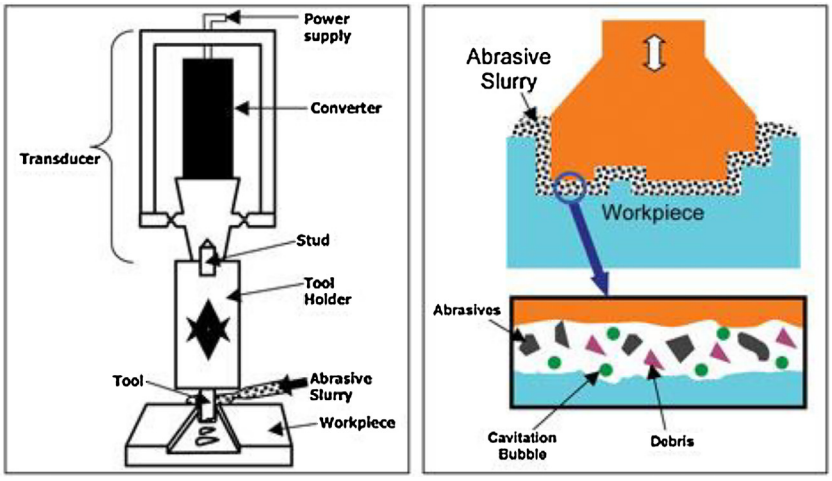


Fig. 26. Principle and working of ultrasonic machining [281].

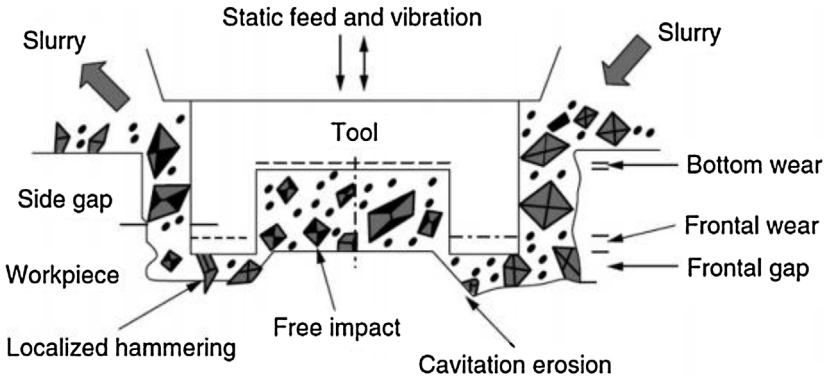


Fig. 27. MRR mechanism in ultrasonic machining [295].

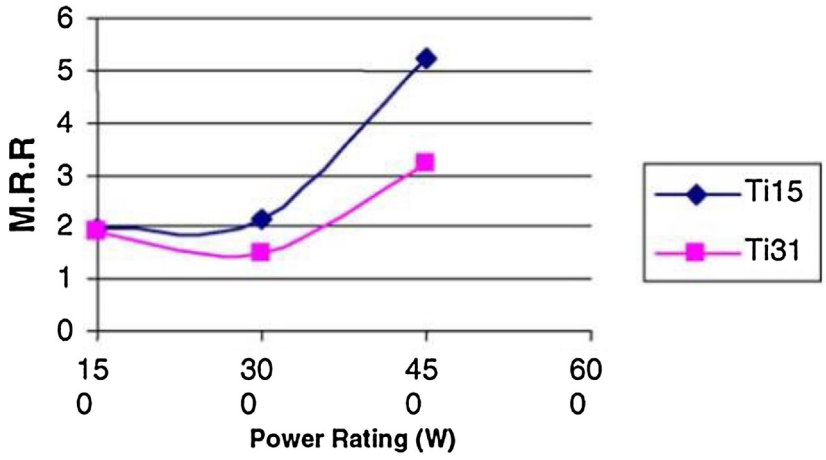


Fig. 28. Effect of ultrasonic power on MRR on Ti15/Ti31 using alumina slurry of 320 grit size and HSS tool [305].

the traverse force (F_x) but the percentage of reduction is lower as compared to that of the axial force. A subsequent FEM analysis supported aforementioned observations. A similar ultrasonic setup as proposed by Park has been designed by Kumar [80], as shown in Fig. 35. This setup was integrated with conventional FSW machine to investigate the effects of ultrasonic vibrations in the friction stir processing (FSP) of aluminium alloy. This comparative study between the ultrasonic assisted friction stir processing (UaFSP) and conventional friction stir processing revealed a substantial decrease in F_z and F_x for different process parameters in the presence of ultrasonic vibrations (Fig. 36). The percentage reduction in axial force was found to be higher than that in the traverse force. He further studies about the effect of UaFSP on mechanical properties. This study also reported improvements in process efficiency (FSP Eff.) and the mechanical properties of processed material such as yield strength (YS), ultimate tensile strength (UTS), elongation (ELNG), and micro hardness at rotational speed 1200 rpm and processing speed 40 mm/min (Fig. 37). For a similar mode of ultra-

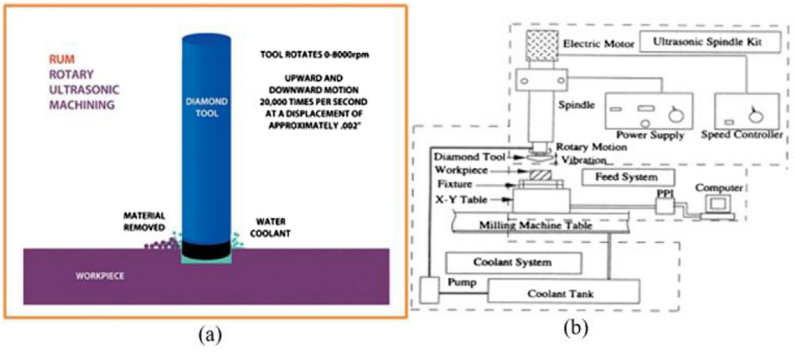


Fig. 29. rotary ultrasonic machining (a) simple schematic [323] (b) experimental set up [324].

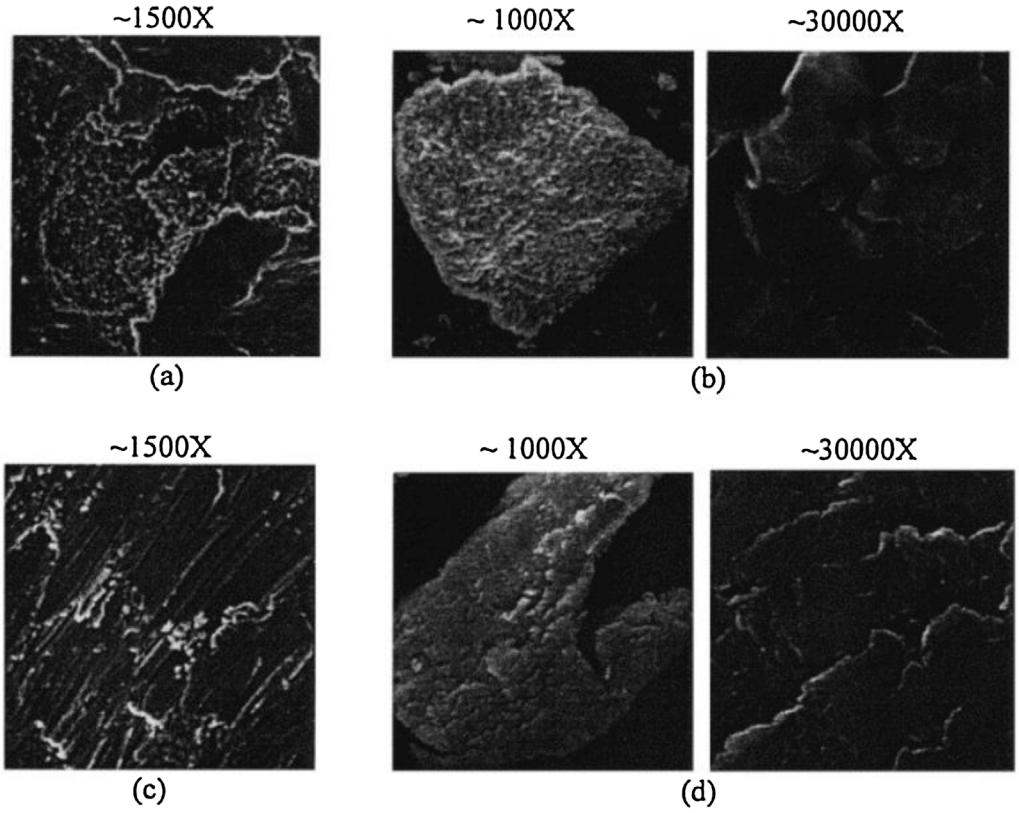


Fig. 30. Two modes of material removal in rotary ultrasonic machining of ceramics. (a) The machined surface generated by brittle fracture mode. (b) The chip generated by brittle fracture mode. (c) The machined surface generated by ductile mode. (d) The chip generated by ductile mode [325].

sonic exertion as Park [226,227,246], Ahmadnia et al. [228] studied the influence of UaFSP parameters on mechanical and tribological properties. They found that application of ultrasonic vibrations improves the grain distribution and mechanical properties.

They suggested that ultrasonic vibration power is an important factor which has positive effect on both mechanical and tribological properties of weld. Rostamiyan et al. [225], using a UaFSW system similar to Park [226,227,246], conducted friction stir spot welding (FSSW) of aluminiuim alloy. They reported that ultrasonic vibrations in the FSSW improves lap shear force and weld hardness.

3.4.2. *Ultrasonic vibrations exerted in axial direction to FSW tool*

This system permits the transfer of ultrasonic vibrations in the axial direction to FSW tool as shown in Fig. 38. The advantage of this system is that it is easily integrated without use of any bearings, allowing direct transfer of vibrations in the desired location.

Ruilin et al. [229] applied such type of ultrasonic integration and proposed that the impact of the vibrations on the temperature field of UaFSW is less effective while working at lower welding speeds. This was attributed to the additional heat input generated by the ultrasonic vibration in the UaFSW model. At lower welding speeds, the maximum temperatures of both processes were identical but at higher welding speeds, decrease in the maximum temperature is more in FSW than in UaFSW. Their numerical simulations showed a relatively larger welding process window with UaFSW. Ma et al. [230] observed that the mechanical properties such as tensile strength, hardness and weld elongation of welds in UaFSW improved when the supply of ultrasonic energy was 50%, however, the hardness and tensile strength were in a state of decline when the ultrasonic energy reached beyond 50%. Amini et al. [232] found that ultrasonic vibrations, applied through the FSW tool, can considerably reduce axial force of FSW due to improved penetration [80,231,232]. Their investigations further revealed that increase in

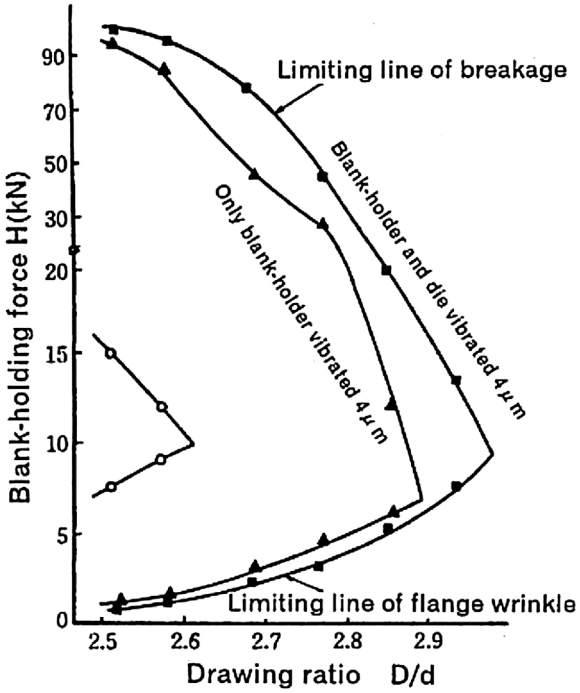


Fig. 31. Effect of ultrasonic oscillation on blank holding force and drawing ratio [351].

the welding speed of FSW increases the downward force slightly while the reduction of rotation speed decreases the downward force. The above study also made evident that ultrasonic vibrations increase the temperature fields due to improved stirring of plastic material. UaFSW enhanced the joint strength and elongation by ~10%. Later on Amini et al. [231] designed and developed the bending vibrational tool to supply the ultrasonic vibrations in bending mode.

When compared to a simple FSW tool, the bending vibrational tool improved the FSW process by lowering the welding forces and increasing the process temperature, as shown in Figs. 39 and 40.

3.4.3. Ultrasonic vibrations exerted to the workpiece ahead of the FSW tool

Owing to the development of effective systems for integration of ultrasonic vibration in to the FSW machine, More recently, Wu et al. [233–239] developed a novel ultrasonic system or sonotrode and integrated it to FSW machine through simple linkages. The entire system, as shown in Fig. 41, is referred to as ultrasonic vibration enhanced friction stir welding (UVeFSW). In the UVeFSW, ultrasonic vibration was directly and dynamically transmitted along the joint line of the workpieces just ahead of the FSW tool by the

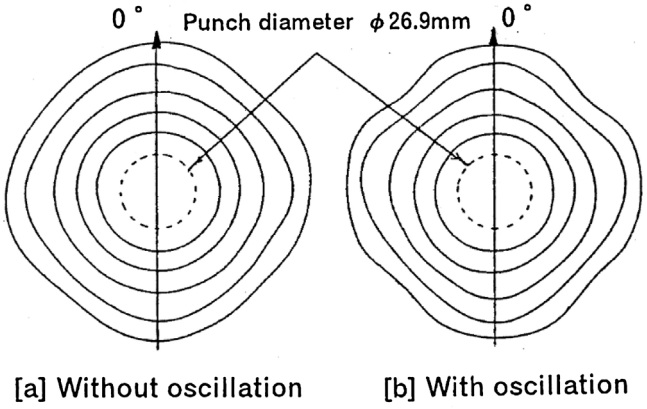


Fig. 33. SPCE sheets developed after deep drawing; drawing ratio D:d 2.68 [351].

sonotrode during welding. In a series of investigations comprised of experimental studies, microstructure analysis and process modelling on UVeFSW, they examined the effect of ultrasonic vibrations on the temperature, process forces, material flow, weld formation, microstructure and mechanical properties of the welded joints of aluminium alloys [237,239,240,247].

Liu et al. [249] in their study revealed that the improved weld physical and mechanical properties in the UVeFSW process is due to increased material deformation and reduced heat affected zone, respectively. The weld microstructures revealed that the ultrasonic vibration enhances the material flow, deformation in the weld nugget zone (WNZ) and widens the thermo-mechanically affected zone (TMAZ). Because of the enhancement of material flow, the microhardness of the WNZ and the tensile strength of the joints increase to some extent. Liu and Wu [239] performed material flow analysis in the UVeFSW of aluminium alloy by using marker insert and sudden stop action technique. They showed that the improved material flow in UVeFSW is due to the additional material softening due to the ultrasonic vibrations. Shi et al. [241] modelled the quasi-steady state of the UVeFSW process. The model revealed that the improvement in material flow is because of the ultrasonic-induced increase in material flow velocity and reduction in flow stress. The above UVeFSW model [241] also revealed that in the angled mode of exertion the ultrasonic vibrations have little influence on the net heat generation and process temperature during welding. Further, they updated the model by associating it with a constitutive equation that took into account the acoustic softening effect [242]. The improved version of UVeFSW model revealed that ultrasonic energy density is an important parameter in the UVeFSW process. Higher is the ultrasonic energy density, more is the reduction in activation energy for deformation [243]. Zhong et al. [248] investigated the effects of UVeFSW on process load, temperature and material flow in UVeFSW. They observed that the

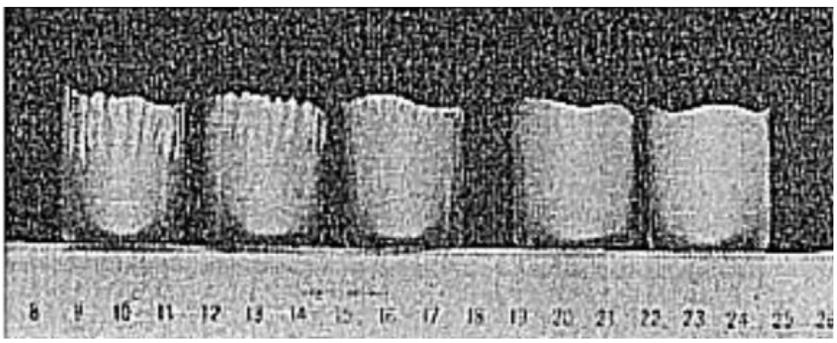


Fig. 32. Elimination of wrinkles of deep drawn cups due to Oscillation [351].

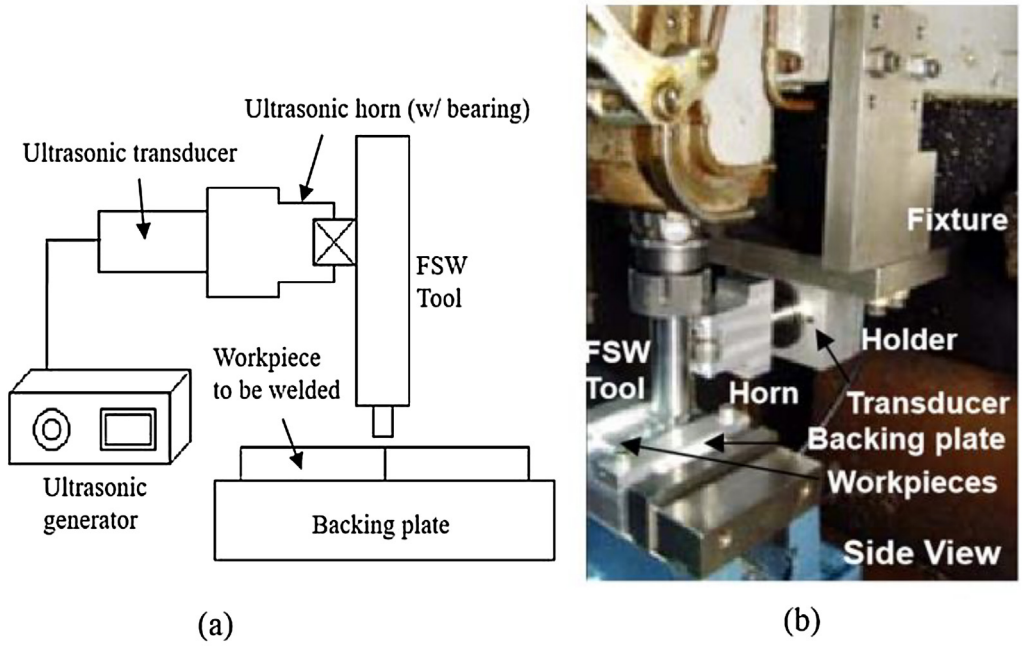


Fig. 34. Integration of ultrasonic vibrations in horizontal direction to FSW tool (a) Schematic (b) actual integration into the machine [246].



Fig. 35. (a) Ultrasonic horn and (b) ultrasonic setup assembly [80].

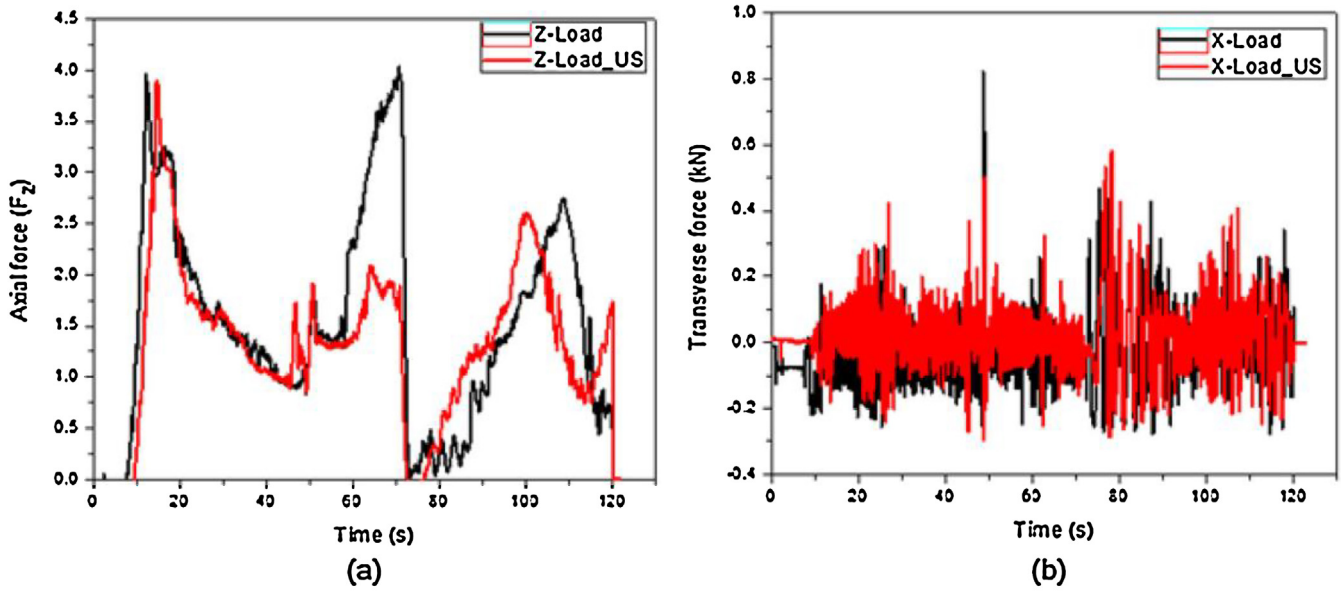


Fig. 36. (a) Axial force and (b) transverse force for FSP and UaFSP at 1200 RPM and 10 mm/min [80].

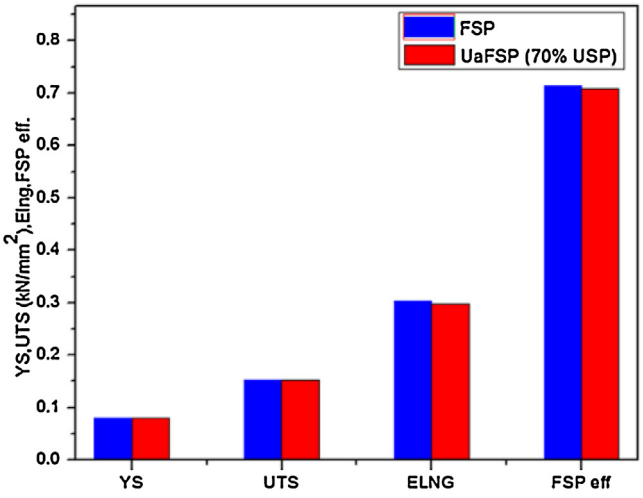


Fig. 37. YS, UTS, ELNG and FSP Eff. comparison of FSP and UaFSP at 1400 rpm and 40 mm/min [80].

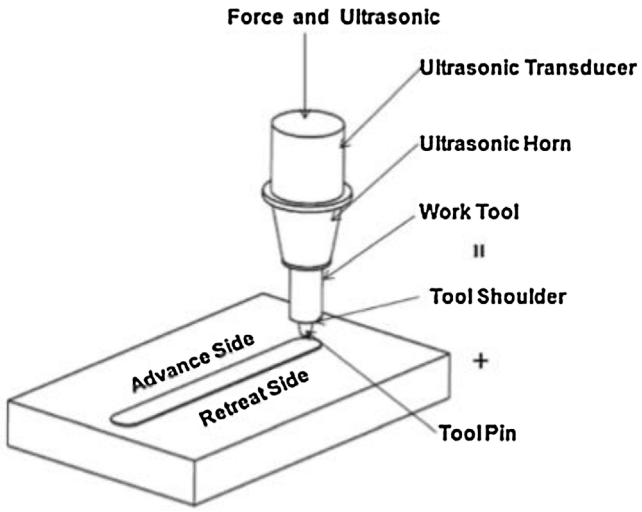


Fig. 38. ultrasonic vibrations along axis of FSW tool [229].

ultrasonic vibrations in FSW lower the traverse force, tool torque and axial force. However, they found that the actual thermal effects of ultrasonic vibration were not propagated into the UVeFSW process, as shown in Fig. 42. Moreover, their study on dissimilar aluminium welding by UVeFSW showed improvement in material flow on either side of the weld line and suppression of inner weld defects in UVeFSW, as shown in Fig. 43. Complete elimination of tunnel defect by ultrasonic application in FSW was observed by Liu et al. [240]. Padhy et al. carried out a comparative study of microstructure evolution of aluminium alloy weld nugget in FSW and UVeFSW using the advance orientation imaging microscopy [244]. Their studies revealed that ultrasonic vibrations improved the grain refinement and subgrain formation in the center of weld nugget. Also, the weld nugget center exhibited a grain structure with maximum amounts of completely-recrystallized and under-recrystallized grains. Gao et al. followed with the observations that ultrasonic vibrations caused higher strain induced grain refinement in the weld nugget center [245], indicating an increasing in strain by ultrasonic vibrations. A combination of the above observations

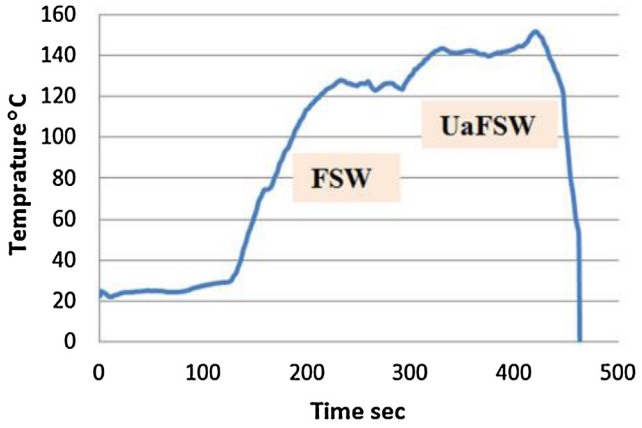


Fig. 39. Temperature variations of tool shoulder with and without bending vibrations [231].

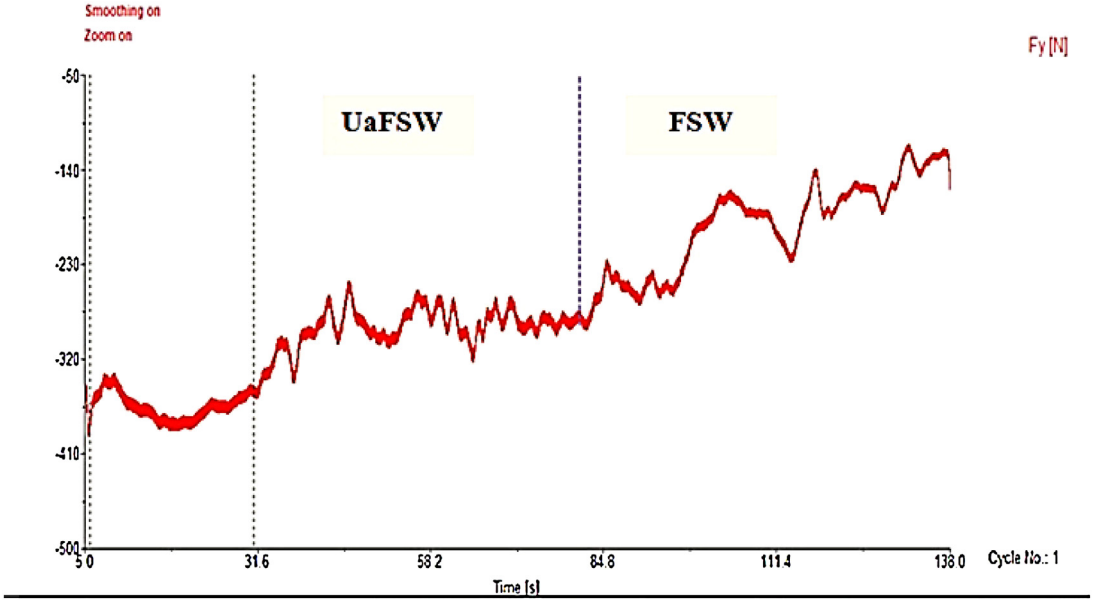


Fig. 40. Welding force on Teflon plates along feed direction with and without vibrations [231].

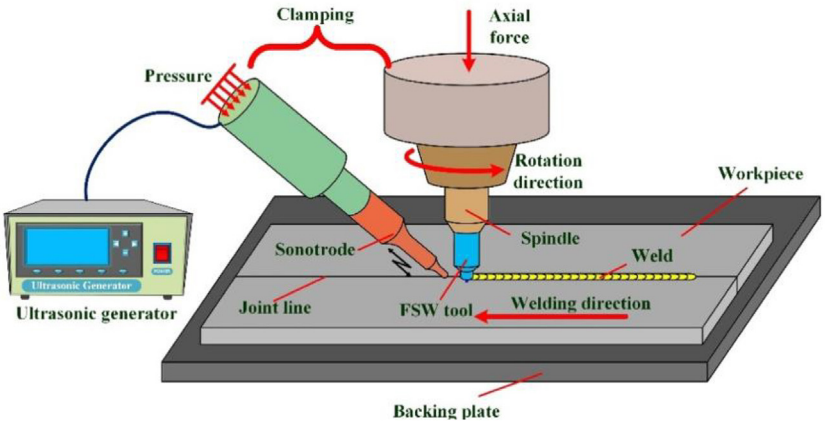


Fig. 41. Schematic of UVeFSW [248].

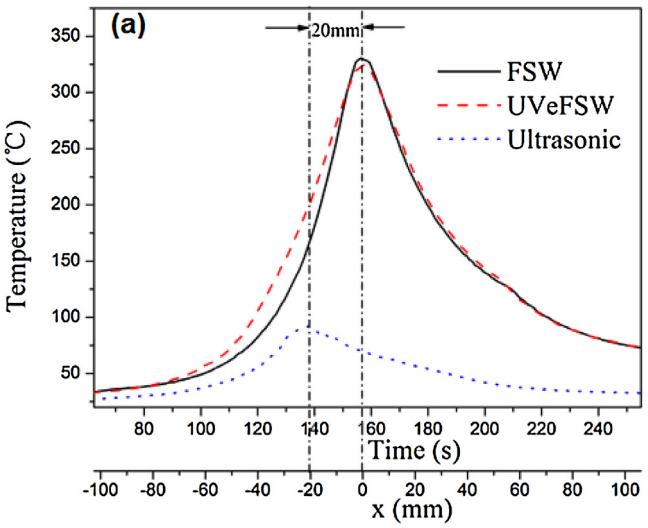


Fig. 42. Thermal cycles of FSW, UVeFSW and ultrasonic treatment processes (400 rpm, 65 mm/min [248].

suggested that the effect of ultrasonic vibrations in the welding of a 6 mm aluminium alloy was maximum at the center and decaying towards the boundaries of the weld nugget, as shown in Fig. 44. Finally, ultrasonic assistance in FSW proved to be robust, safe and superior in terms of process effectiveness and weld properties as compared to conventional FSW. Application of ultrasonic vibrations in FSW improved mechanical working and material flow in the welded/processed zone, widened the process window, lowered or eliminated the weld defects and enhanced weld mechanical properties and microstructure. However as the welded materials exhibit different metallurgical and mechanical properties compared to the base metal, proper selection of process parameters for further processing is very important.

UVeFSW further could be implemented to successfully join and hard metals such as titanium and hardened steel. Joining of wood pieces are still under initial stages. An optimized design of ultrasonic components as well as direction of application enhance the energy density and lowers the ultrasonic losses. FSW is recently adopted by electronic manufacturing company 'Apple' for his iMac monitor to join two sheets which slimmed it (iMac monitor) by 40%.

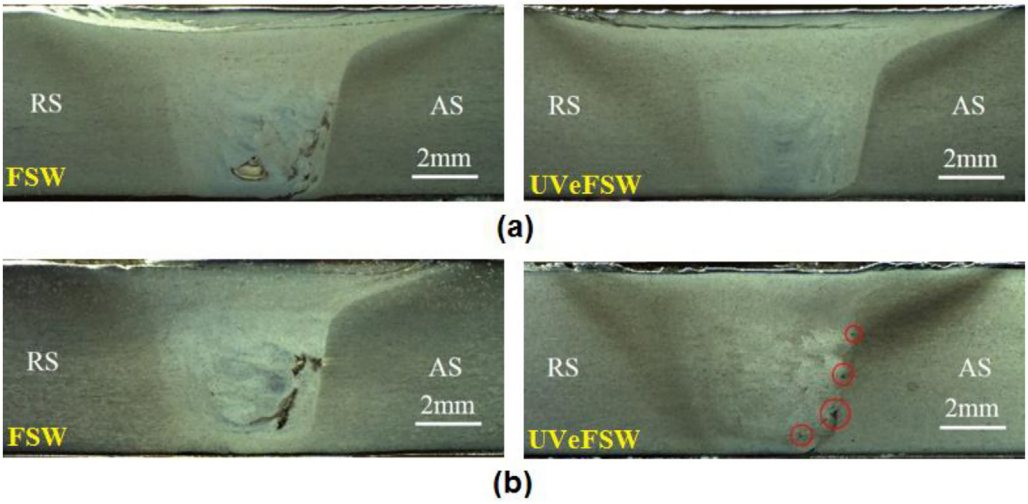


Fig. 43. Cross-sectional macrographs of AA2024-T3 FSW butt joints with and without ultrasonic at welding parameters (a) 200 rpm, 110 mm/min, Z = 0.1 mm, and (b) 400 rpm, 330 mm/min, [248].

4. Application of ultrasonic vibrations in machining process (ultrasonic machining)

Machining is considered as an inevitable metal working operation. Machining is a term that is primarily employed to describe different sets of material removal processes which generally employ a cutting tool that is intended to removes unwanted material from a workpiece to provide it a predefined shape. Machining is also very commonly termed as a secondary manufacturing process, where a very small quantity of material is intended to be removed and the machining time is quiet short.

Application of ultrasonic vibrations in small scale metal-cutting operations started since the early 1950s [250–252] by the name ultrasonic impact grinding or USM [28,38,44,253]. Use of ultrasonic vibrations in machining process and development of dedicated USM technology was aimed purely at overcoming the difficulties associated with the machining of hard metals [254]. In other words, the purpose of ultrasonic application in the machining of hard materials is to achieve necessary force reduction and good finishing of the end product [255–258]. Early development of USM was relied on a novel system that allowed conversion of low frequency electrical energy to a high-frequency electrical signals. The high frequency oscillations were transferred onto the machining tool via a transducer [27,28,37,254,259]. Machining processes coupled with ultrasonic vibrations hold several advantages over the conventional machining processes [260–262]. USM is considered as a low material removal process where mirror image of specially configured tool is created in hard and brittle material. USM is suitable for hard and brittle materials such as glass, piezo ceramics, carbide, alumina ferrite, sapphire and ceramic matrix composites etc. USM is usually a non-thermal, non-electrical as well as non-chemical process.

4.1. Ferrous and nonferrous alloys

USM is a general reference to several processes such as ultrasonic abrasive machining, ultrasonic drilling, ultrasonic cutting, ultrasonic dimensional machining, slurry drilling etc. The USM (turning) is a more efficient method than the conventional turning in terms of cutting force, cutting stability, chip formation, tool blunting, tool wear, surface finish, etc. Furthermore, the USM process offers technological feasibility for the machining of hard materials such as nickel and titanium alloys, hardened steels, ceramics, and tungsten carbides, etc. [263–265]. In addition, machining processes integrated with ultrasonic vibrations enhance the productivity owing to the reason that these processes save the ~5–10% of machining time as well as 30% of machining cost [266]. Several investigations on USM based on theoretical analysis [267–269], finite element simulation [270] and experimental results [47,271,272] summarized that ultrasonic vibrations induce reduction in cutting force by lowering the friction at tool-workpiece interface [272] as well as due to the pulse cutting characteristic of the vibratory cutting tool [266–268,273]. The USM/cutting is reported to be more efficient while working at low cutting speed [264–266]. On the other hand, the mechanism of cutting at tool–workpiece interface is dependent on some factors such as cutting speed, ultrasonic vibration frequency and amplitude. Fig. 24 shows schematic of ultrasonic vibration tool integrated with the turning process [274].

Fig. 25 depicts the variations in flank wear and cutting force components in radial (F_r), tangential (F_t) and axial (F_a) directions with respect to the cutting speed after 10 min of cutting during conventional turning (CT) and ultrasonic vibration turning (UVC) [274]. Approximately 68–85% reduction in cutting force and significant reduction in tool flank wear was observed for a cutting speed of 5–20 mm/min after applying ultrasonic vibrations. The reduction in cutting force was seemed to be higher at lower cutting speeds. The above reductions are attributed to the reductions in the interfacial friction and built up edges which were continuously removed by the ultrasonic vibrations in the midst cutting. Fig. 26 shows basic schematic and working of an USM process. As the tool does not make physical contact with the workpiece, therefore, a very small downward force is required to do machining.

USM is employed to both metallic and non-metallic materials, specifically, to materials with low ductility [43,275] and hardness value above 40 HRC [20,21,68,253] such as inorganic glasses, silicon nitride, nickel/titanium alloys, etc. [37,46,276–278]. Besides, the heat generation is negligible, USM of extremely hard and brittle materials such as glass, diamond, and ceramics is technologically feasible and prone to be more effective. In recent years, ultrasonic vibrations are extensively used to develop systems that produce precision microstructures, i.e., microstructures similar to that of copper or aluminium, in steels and optical materials with complex geometries and surface finishes [279]. Ultrasonic vibrations have enabled the machining of precision components i.e., components with close dimensional accuracy, also achieving higher tool life during cutting of hard materials such as polycrystalline diamond and conventional carbide tools [280].

Chang and Bone [282] investigated the effect of ultrasonic vibrations on chip formation and thrust forces. They observed that the height and width of burr were notably reduced using ultrasonic assistance. They also observed a reduction in thrust force during the same. Ishikawa et al. [283] reported the effect of combining low frequency ultrasonic vibration at the core drill in the drilling operation. They focussed on proposing a relationship between drilling force and hole accuracy. They observed that drilling force was decreased in the presence of ultrasonic vibrations and further was reduced with increase in ultrasonic vibration amplitude. Babitsky et al. [264] experimentally investigated the effect of ultrasonic vibrations on the turning of aviation materials such as C263, Inconel-718 and mild steel. They found that the magnitudes of roughness profiles are reduced by nearly 50% in the case of Inconel-718 and C263, whereas the reduction was even more in mild steel. With the application of ultrasonic vibration, the regularity of roughness profile is improved considerably and the surface became smoother along the axial direction as the conventional cutting process is transformed into high frequency vibration impact process. This transformation increased the dynamic stiffness of the lathe-tool-workpiece system and improved the accuracy of turning. In the case of mild steel, the improved surface roughness was attributed to elimination of built-up-edges at low cutting speeds during ultrasonic vibration. Similar results have been reported by Skelton et al. [284] and Kim et al. [265].

4.2. Material removal mechanisms

Computation of metal removal rate (MRR) during machining of metals is very crucial for economic machining. Several researchers have developed numerous methods and models [21,285–288] to optimize and increase the MRR. Some of the proposed mechanisms of material removal in machining processes are direct hammering of abrasives [26,286–288], microchip removal [68,288,289], cavitation effect [38,254] and chemical abrasion [38,254]. The formation of cavitation during USM alarms that the cavitation bubbles were attributed to the high pressures generated by the 18–22 kHz oscillations on the workpiece surface where subsequent collapse of the bubbles led to material erosion towards the impacted locations [290,291]. As obvious, this impact was more significant when abrasives of higher grain size were employed. The cavitation bubble formations facilitate further crack propagation in the material which is most significant for material removal during USM. The same concept is also applicable for USM of ceramics and composites [292,293]. Ultrasonic vibrations enhance the velocity and

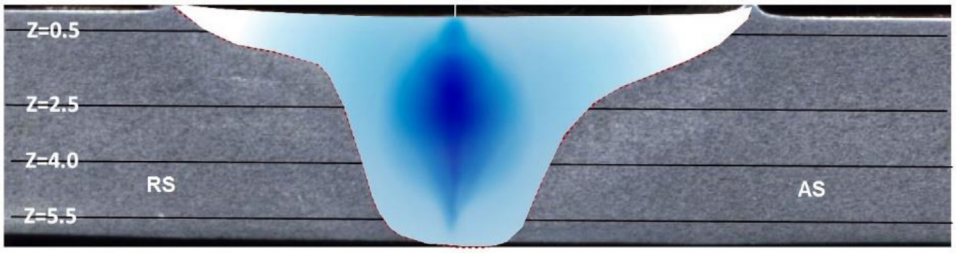


Fig. 44. Schematic of the degree of ultrasonic effect in the weld nugget during UVeFSW [244].

turbulence of the sharp edged abrasive and render an inelastic deformation which develop the crack on material surface. Further increment in load cause propagation of the cracks. The cracks begin to close during an unloading, leading to leaving space for the voids [287,293,294]. Increase in the vibrations frequency (up to 400 Hz) increase the MRR very rapidly (Fig. 27). MRR varies linearly with further increase in frequency, between 400 Hz–5 kHz [21,52,295]. It depicts that the ultrasonic vibrations are helpful to increase MRR only up to 5 kHz. Beyond that, MRR decreases rapidly, indicating that the ultrasonic vibrations are no more beneficial [296]. It was reported that MRR is also a function of the abrasive size. Therefore, optimization of abrasive size is another vital aspect to have economical MRR [297,298]. Beside, MRR is also influenced by tool size, shape and mean grit size [39,59,299]. An optimum size enhances its life [118,288,298]. It was reported that MRR increases with amplitude of vibration. On the other hand, to achieve optimum cutting rate, oscillation amplitude should be equal to the abrasive grit mean diameter [56,277,296]. Therefore an optimum amplitude is crucial to enhance MRR and the machining process. Several instruments and methods are in existence to measure the amplitude of ultrasonic vibrations and are detailed elsewhere [41,44,277,300].

4.3. Ultrasonic machining of titanium and its alloys

At present, hard materials such as titanium and its alloys are extensively used in different manufacturing application owing to their strategic importance as unique lightweight, high strength alloyed structurally efficient metal for critical, high-performance aircraft [301]. Therefore, machining of these materials has become vital. However, the conventional machining process are not capable of machining these hard materials [302]. Some advanced processes such as electric discharge machining (EDM), electric chemical machining (ECM) were developed to machine hard metals at good MRR. However, surface finish of machined material continues to be an issue [301,303,304]. Studies have proposed that smooth machining of hard metals with sound surface finish is feasible by combining ultrasonic vibrations to conventional machining under similar process conditions [21,68,78,253,277].

A wide spectrum of investigations has been reported that focus on improving material removal during USM of hard and brittle metals [26,285–288,301,305] in terms of microchips [68,288,289], cavitation effect [26,277] and chemical abrasion [38,254] etc. Fig. 28 shows the effect of ultrasonic power on MRR during drilling of Ti15/Ti31 by using alumina slurry of grit size 320 with HSS tool [305]. It can be noted that while drilling large size holes, the machine power requirements can be reduced by trepanning the shape with a hollow tool rather than using a solid tool [306]. The tool hardness play an important role in machining these hard metals; the tool must be harder than that of workpiece to be machined. The choice of tool material should be carefully made to prevent tool wear during machining of the harder and tougher Ti-alloy [307]. The order of tool materials in terms of their application feasibility and superiority [33,308] is as follow: Nimonic 80A > thoriated tungsten > silver steel > stainless steel > maraging steel > titanium > mild steel. Tool wear is very important parameter needed to be considered in the USM because it a wide effect on MRR [33,309]. Studies reported that tools with diamond tip exhibit good material removal characteristics and extremely low wear rate [21]. Two types of complex tool wear patterns are found duirng USM, longitudinal wear [284,310] and lateral wear [311–313]. Venkatesh [314] found that tool wear was maximum at a static load that is optimum to achieve maximum MRR [285]. During operation, if the hardness of the tool increases by work hardening, the penetration of abrasive grains into the tool would decrease, resulting in higher work piece MRR. It is reported that the hardening of tool is maximum at the periphery and minimum at the center [315]. Metals softer than the workpiece can't be used as tools materials in USM because they would generate burrs at large oscillation amplitudes [44,296].

4.4. Machining of ceramics

Ceramics are non-metallic, inorganic solid materials compromising of the strong ionic and covalent bonds. They exhibit good thermal and electrical insulation properties. Now a days, ceramic materials are used in various industrial applications owing to have superior properties than metals. Some major applications of ceramics are as gas burner nozzles, biomedical implants, mechanical bearings, space shuttle tiles and ballistic protections etc. Ceramics are also proving their wide applicability in several automotive components [274,316–318]. However, their application is restricted owing to the high cost of fabrication. Hence, development of cost effective methods for the machining of ceramic materials are essential and need a great attention. Some processes are evolved to do machining of ceramics such as rotary ultrasonic machining (RUM) process, laser processing, electric discharge machining (EDM), etc. Among these, RUM process is proved to be more effective, relatively cheaper and robust under the traditional machining environment. Therefore, RUM is used for most of the industrial scale machining of ceramics.

RUM was invented in 1964 at United Kingdom Atomic Energy Authority [319,320]. Fig. 29 shows the schematic and experimental set up of RUM, respectively. In the RUM, a rotating core drill with metal tool-bonded diamond abrasive is ultrasonically vibrated in the direction of its axis, while the spindle is fed toward the workpiece at a constant pressure. A coolant is supplied through the drill core to cool it. The coolant also pumps and washes away the derbies, which otherwise would lead to jamming of the drill. Due to the tool-bonded abrasive and simultaneous rotation and vibration, RUM provides a fast and high-quality machining of a variety of glass and ceramic materials. Controlling of RUM is done by optimizing the process parameters such as applied static pressure and rotational speed as well as ultrasonic parameters such as frequency, amplitude and abrasive parameters such as diamond type, size and concentration. To produce the best machined components by RUM, the above parameters must be optimized. Details of the machining parameter optimization methods are available

elsewhere [56,321,322]. Modelling studies on the RUM of ceramic materials have proposed two modes of material removal, brittle fracture mode and ductile fracture mode [294].

Two types of brittle fracture models are reported for the RUM process in past studies. The first model, which is theoretical and based upon the indentation fracture theory. It is able to fairly explain the material removal process [325]. The second model uses a mechanistic parameter to determine MRR. The MRR calculations show good agreements with experiment results in case of the RUM of magnesia stabilized zirconia [326]. Fig. 30 shows the two modes of material removal in RUM during machining of ceramics.

5. Application of ultrasonic vibrations in metal forming process (ultrasonic forming)

Metal forming, also known as metalworking is a process of devising parts of the metal and objects by applying mechanical deformation. In this process, without any loss of material or change in mass, the shape of the workpiece is altered. It is evident that metal forming has more uniform characteristics across its sub processes as compared to cutting and joining. Depending on the nature of stress applied, the forming process are different types, such as compressive forming (rolling, die forming, extrusion, forging, indenting), tensile forming (stretching, recessing, expanding), combined tensile and compressive forming (deep drawing spinning flange forming) and bending shearing etc.

Metal forming is an important stream of metal working where ultrasonic vibrations are extensively used to improve material properties. The forming process integrated with ultrasonic vibrations is called ultrasonic forming (UF). UF comprises of several stages such as stress superposition [327], softening effect [328] or hardening effect [329], change in coefficient of friction [27] and friction vector effect [330,331]. Rosenfield et al. [332] integrated radial ultrasonic vibration to the blank-holder and die in deep drawing process. They found that vibration with limited frequency affect the deep drawing ratio while the ratio can be varied by altering the ultrasonic frequency. They also reported that wrinkles formed during forming can be eliminated by increasing the vibration amplitude. Severdenko and Klubovich [333] pointed out that the effect of ultrasonic vibrations as analogous to the rise in temperature, which induce considerable reduction in the rate of work hardening. Langenecker performed experimental studies and inferred that outcomes of UF can't be explained by acoustic stress alone [6,332]. Further, he discussed various possible mechanisms and suggested that at low energy level, the superposition mechanism prevails while that of at high energy level, overheating of the sample causes softening. Moreover, "volume effects" on UF were studied for different frictional characteristics [334,335]. It was seen that after applying vibrations in three orthogonal directions, the friction was reduced in all cases. Also, it was observed that changes in friction coefficient in a wedge test could find a 40% reduction in dry friction and a 20% reduction under lubricated conditions [334,335]. Overviewing the benefits of ultrasonic vibration in the forming, it became very popular for other industrial applications especially, forging, extrusion and drawing. Addition of ultrasonic vibrations resulted in considerable improvement in the forming conditions. During an early stage, Severdenko and Klubovich [333] studied the effects of ultrasonic vibration on the forging of aluminium alloy. Their observations such as reduction of forging force to zero, elimination of "barrelling" and reversal of residual stress distribution were the first of their kind. They suggested that such results would be due to the reduction in friction as well as thermal softening. Similar results were also reported by different researchers [336–338], who observed considerable reduction of force with reference to superposition of amplitudes. Moreover, ultrasonic vibrations with higher amplitudes produce good results owing to propensity to cause higher thermal softening.

Application of ultrasonic vibrations in the extrusion process has also proved beneficial, leading to improvement in various mechanical properties. Author achieved a force reduction of significant value by applying ultrasonic vibration [339]. He also observed significant variations in microhardness and yield stress. Jones [340] performed ultrasonic integrated extrusion on lead and aluminium alloy. He found that, by vibrating the die, the force could be decreased by 15% while the extrusion rate could be enhanced significantly by 88%. These improvements were attributed to the lowering in friction due to the ultrasonic assistance. Similar observations were made by Rosenfield [332], who reported a force reduction in the range of 15 and 30% by additional ultrasonic interference. Some researchers [332,341,342] integrated ultrasonic vibrations in wire drawing operations. In such operation, the ultrasonic vibrations are usually added to the die along the wire axis. They reported significant reduction in wire drawing speed, up to 65%, leading to small decrease in microhardness. Jones integrated ultrasonic vibrations to the drawing process to realize, not only a substantial reduction in drawing force but also an improved surface finish [340]. Pohlman et al. and Winsper et al. conducted similar studies to find a drop in mean peak stress value [334,343]. They concluded ultrasonic integration alone did not have much effect on the peak stress value, which meant that the drop in mean stress was because of acoustic stresses superposition. More advanced application of ultrasonic vibration in forming was carried out by Winsper and Sansome [344,345] who used three dies while the ultrasonic vibrations were supplied to central die. By having that, they claimed a reduction in draw force by ~50 lbf. This was attributed to the lateral compression of workpiece, thereby utilizing full amount of vibrations for the softening. In similar studies, Oelschlagel [346] also reported reduction in force by 37%–62%. Jones [340] tested axial-mode dies in plug drawing and reduced the draw force by up to 80%. Author [347] used radial die vibrations with a plug bar of tuned length (so that axial vibrations would be induced in the plug) to achieve a force reduction ranging from 10 to 30%. Biddell and Sansome [348] in their experiments, designed and manufactured different variety of radial-mode dies so that they could obtain improved depths of draw, mainly while working at low speeds. They performed several trials on this experimental setup and concluded that the effect was a result of reversal of the friction vector. They also concluded that the draw depth increase of ~12% can't be achieved at every time by using the radial dies [349]. Tisza [350] used a complex combined axial-radial mode die so that the axial vibrations at the transducer could be converted to radial at the die. He used a 15 μm amplitude from the transducer end to observe an increase in deformation by up to 15%. The reason behind the improved deformation was the reduced "friction factor", friction vector reversal and a modified tensile strength under the ultrasonic action. Jimma et al. [351] studied an ultrasonic assisted deep drawing process where a blank-holder or die plate was vibrated in a radial mode. They concluded that the limiting drawing ratio (LDR) was considerably increased with decreasing blank holding force in the presence of ultrasonic vibrations, as shown in Fig. 31. They also noticed disappearance of wrinkles after applying ultrasonic vibrations, as in Fig. 32. A developed blank after deep drawing is presented in Fig. 33, over which, the circles are etched. It was seen that the ears appear in the directions of 0° and 90° against the rolling direction. The ears are more evident while applying the ultrasonic Therefore, it is seen that when ultrasonic vibrations are applied in the forming processes, almost all the applications show improvements in the form of either reduction in the forming force or increase in the apparent formability of workpiece. These improvements have also been further investigated to advance the theoretical base and as a result, several theories have

come into existence to lucidly explain the observations. Despite these, widespread use of ultrasonic vibrations in the industries is yet far from practicality. In the 1970s, some review articles suggested that full production status was imminent [352,353] while Biddell and Sansome [354] shifted the blame on the industries for not making use of the technology. After almost half a century, the status of ultrasonic application in the industries is still unchanged although there has been great progress in its research and related developments. However, the industrial fraternity would probably defend by saying that integration of ultrasonic vibrations increase cost of production while the benefits are not enough so that the extra cost and complication of the ultrasonic equipment could be justified. However, high-power ultrasonic vibrations are extensively used and applied by section of industry such as welding and machining.

From the above discussions, it is clear that ultrasonic vibrations are employed both solely and as an assistant in the welding of ferrous and nonferrous alloys. When applied solely in the ultrasonic welding process, joining of a wide variety of similar and dissimilar metals has been realized. The resulted welds show good integrity with respect to their physical and mechanical properties. Ultrasonic vibrations have been applied as assistants in both fusion and solid state welding processes. When assisting the arc welding processes, ultrasonic vibrations bring about improvements in the welding process window, various process variables such as the arc features, droplet transfer frequency, weld penetration depth, weld bulk, weight ratio etc., mechanical properties and microstructure in terms of grain refinement. When accompanied with the FSW process, the ultrasonic vibrations cause either sizable or feeble increase in process temperature based on the mode of ultrasonic application. Irrespective of the application mode, the assistant ultrasonic vibrations improve the material flow and widen the FSW process window. Also, these vibrations induce improvements in the weld physical and mechanical properties as well as in the weld microstructure. The FSW microstructure with ultrasonic vibrations exhibits higher deformation and finer grain structure. Hence, assistance of ultrasonic vibrations both in arc welding and FSW improves the process effectiveness and resulting weld properties. Therefore, it is reasonable to arrive at a unanimous assertion that ultrasonic vibrations serve towards the benefit of the welding processes, weldments and ease of production. This presents a fair case of their further development with a goal for their industrial deployment. Apart from their application in metal joining, there are also other areas of metal processing such as machining and forming where the role of ultrasonic vibrations is significant.

6. Conclusion

In a nut shell, it can be depicted that ultrasonic vibrations do play a substantial role in the welding and metal processing. There is tremendous scope for application of ultrasonic vibrations for establishing cost effective manufacturing solutions for relatively tough and ductile metals such as titanium, nickel-alloys. It is quite obvious that ultrasonic vibrations when employed in the welding and other manufacturing processes, yield enormous benefits in terms of improved mechanical properties, adept surface quality, less tool wear, improved material flow, enlarge process window, improved MRR, and uniform grain growth etc. From the economics point of view, it is acquit that due to improvement in several aspects of ultrasonic vibrations assisted processes the overall cost of the finished product is quiet reduced. However there is still wondrous scope for further research in many crucial aspects of ultrasonic vibrations assisted process and its variants such as assessment of cavitation phenomenon by using dynamic simulations, delamination of drilled holes, sub-surface damage, process optimisation, complex vibration systems in ultrasonic forming and welding, morphological study of welded and machined parts, improved and optimized design of weld reed system and optimized integration of ultrasonic vibration in the FSW. Numerical modelling of various aspects of ultrasonic assisted processes is another crucial area to be looked into. It was seen that these aspects do not get proper attention and in-depth consideration by different researchers and needed to be further amplified by suitable research and developments methodologies. In addition, the current investigations with these variants are mostly limited to soft metals such as aluminium. The application of ultrasonic processes for individual variant to harder and complex materials is yet its initial stage of exploration and needed a great aid. In addition to the experimental investigations on process variables, modelling of the different aspects of each variant as well as texture analysis at different grain orientations has to be carried out to reveal the coupled interaction mechanism of the ultrasonic energy.

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